
KTU
NOTES

MODULE 4

SWITCH GEAR

Circuit Breaker

Introduction: During the operation of power system, it is often desirable and necessary to switch on or off the various circuits (e.g., transmission lines, distributors, generating plants etc.) under both normal and abnormal conditions. In earlier days, this function used to be performed by a switch and a fuse placed in series with the circuit. However, such a means of control presents two disadvantages.

1. Firstly, when a fuse blows out, it takes quite some time to replace it and restore supply to the customers.

2. Secondly, a fuse cannot successfully interrupt heavy fault currents that result from faults on modern high-voltage and large capacity circuits.

Due to these disadvantages, the use of switches and fuses is limited to low voltage and small capacity circuits where frequent operations are not expected e.g., for switching and protection of distribution transformers, lighting circuits, branch circuits of distribution lines etc. This necessitates employing a more dependable means of control such as is obtained by the use of circuit breakers.

A circuit breaker can make or break a circuit either manually or automatically under all conditions viz., no-load, full-load and short-circuit conditions. This characteristic of the circuit breaker has made it very useful equipment for switching and protection of various parts of the power system.

A circuit breaker is a piece of equipment which can

(i) Make or break a circuit either manually or by remote control under normal conditions.

(ii) Break a circuit automatically under fault conditions

(iii)Make a circuit either manually or by remote control under fault conditions

Operating principle:

A circuit breaker essentially consists of fixed and moving contacts, called Electrodes. Under normal operating conditions, these contacts remain closed and will not open automatically until and unless the system becomes faulty. Of course, the contacts can be opened manually or by remote control whenever desired. When a fault occurs on any part of the system, the trip coils of the circuit breaker get energized and the moving contacts are pulled apart by some mechanism, thus opening the circuit.

1. When the contacts of a circuit breaker are separated under fault conditions, an arc is struck between them. The current is thus able to continue until the discharge ceases.

2. The production of arc not only delays the current interruption process but it also generates enormous heat which may cause damage to the system or to the circuit breaker itself.

3. Therefore, the main problem in a circuit breaker is to extinguish the arc within the shortest possible time so that heat generated by it may not reach a dangerous value.

Arc Phenomenon:

When a short circuit occurs, a heavy current flows through the contacts of the circuit breaker before they are opened by the protective system. At the instant when the contacts begin to separate, the contact area decreases rapidly and large fault current causes increased current density and hence rise in temperature. The heat produced in the medium between contacts (usually the medium is oil or air) is sufficient to ionize the air or vaporize and ionize the oil. The ionized air or vapor acts as conductor and an arc is struck between the contacts.

1. The potential difference between the contacts is quite small and is just sufficient to maintain the arc.

2. The arc provides a low resistance path and consequently the current in the circuit remains un- interrupted so long as the arc persists.

3. During the arcing period, the current flowing between the contacts depends upon the arc resistance. The greater the arc resistance, the smaller the current that flows between the contacts.

The arc resistance depends upon the following factors: Degree of ionization, **Length of the arc**, **Cross-section of arc**.

Principles of Arc Extinction:

Before discussing the methods of arc extinction, it is necessary to examine the factors responsible for the maintenance of arc between the contacts. These are:

1. Potential difference between the contacts.

2. Ionized particles between contacts.

a. When the contacts have a small separation, the Potential difference between them is sufficient to maintain the arc. One way to extinguish the arc is to separate the contacts to such a distance that Potential difference becomes inadequate to maintain the arc. However, this method is impracticable in high voltage system where a separation of many meters may be required.

b. The ionized particles between the contacts tend to maintain the arc. If the arc path is demonized, the arc extinction will be facilitated. This may be achieved by cooling the arc or by bodily removing the ionized particles from the space between the contacts.

Methods of Arc Extinction (or) Interruption (or) quenching:

There are two methods of extinguishing the arc in circuit breakers viz.

1. High resistance method.

2. Low resistance or current zero method

High resistance method:

In this method, arc resistance is made to increase with time so that current is reduced to a value insufficient to maintain the arc. Consequently, the current is interrupted or the arc is extinguished.

The principal disadvantage of this method is that enormous energy is dissipated in the arc. Therefore, it is employed only in D.C. circuit breakers and low-capacity a.c. circuit breakers.

The resistance of the arc may be increased by:

1. Lengthening the arc

2. Cooling the arc

3. Reducing X-section of the arc

4. Splitting the arc

Low resistance or Current zero method:

This method is employed for arc extinction in a.c. circuits only. In this method, arc resistance is kept low until current is zero where the arc extinguishes naturally and is prevented from restriking in spite of the rising voltage across the contacts. All Modern high power a.c. circuit breakers employ this method for arc extinction.

1. In an a.c. system, current drops to zero after every half-cycle. At every current zero, the arc extinguishes for a brief moment.

2. Now the medium between the contacts contains ions and electrons so that it has small dielectric strength and can be easily broken down by the rising contact voltage known as restriking voltage. If such a breakdown does occur, the arc will persist for another half cycle.

3. If immediately after current zero, the dielectric strength of the medium between contacts is built up more rapidly than the voltage across the contacts, the arc fails to restrike and the current will be interrupted.

There are two theories to explain the Zero current interruption of the Arc:

1. Recovery rate theory (Slepain‘s Theory)

2. Energy balance theory (Cassie‘s Theory)

Recovery rate theory (Slepain's Theory):

The arc is a column of ionized gases. To extinguish the arc, the electrons and ions are to be removed from the gap immediately after the current reaches a natural zero. If the dielectric strength increases more rapidly than the restriking voltage, the arc is extinguished. If the restriking voltage rises more rapidly than the dielectric strength, the ionization persists and breakdown of the gap occurs, resulting in an arc for another half cycle.

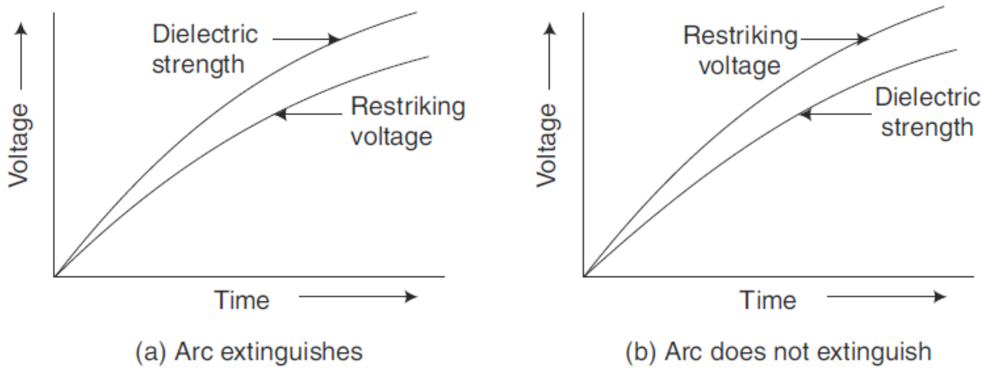
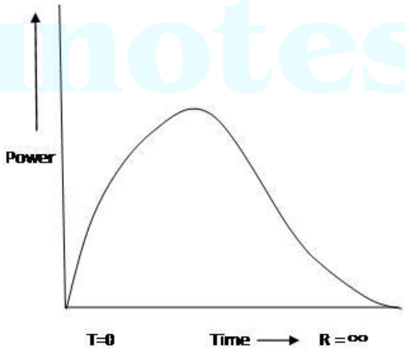


Fig. 14.5 *Recovery rate theory*

Energy balance theory (Cassie's Theory):

Due to the rise of restriking voltage and associated current, energy is generated in the space between the contacts. The energy appears in the form of heat. The circuit breaker is designed to remove this generated heat as early as possible by cooling the gap, giving a blast air or flow of oil at high velocity and pressure. If the rate of removal of heat is faster than the rate of heat generation the arc is extinguished. If the rate of heat generation is more than the rate of heat dissipation, the space breaks down again resulting in an arc for another half cycle.



Important Terms:

The following are the important terms much used in the circuit breaker analysis:

1. Arc Voltage:

It is the voltage that appears across the contacts of the circuit breaker during the arcing period. As soon as the contacts of the circuit breaker separate, an arc is formed. The voltage that appears across the contacts during arcing period is called the arc voltage. Its value is low except for the period the fault current is at or near zero current point. At current zero, the arc voltage rises rapidly to peak value and this peak voltage tends to maintain the current flow in the form of arc.

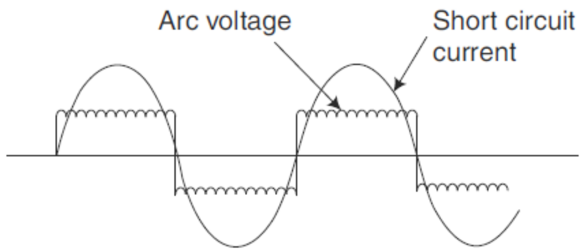


Fig. 14.4 *Short circuit current and arc voltage*

2. Restriking voltage:

It is the transient voltage that appears across the contacts at or near current zero during arcing period. At current zero, a high-frequency transient voltage appears across the contacts and is caused by the rapid distribution of energy between the magnetic and electric fields associated with the plant and transmission lines of the system. This transient voltage is known as restriking voltage .

The current interruption in the circuit depends upon this voltage. If the restriking voltage rises more rapidly than the dielectric strength of the medium between the contacts, the arc will persist for another half-cycle. On the other hand, if the dielectric strength of the medium builds up more rapidly than the restriking voltage, the arc fails to restrike and the current will be interrupted.

3. **Recovery voltage:**

It is the normal frequency (50 Hz) R.M.S. voltage that appears across the contacts of the circuit breaker after final arc extinction. It is approximately equal to the system voltage.

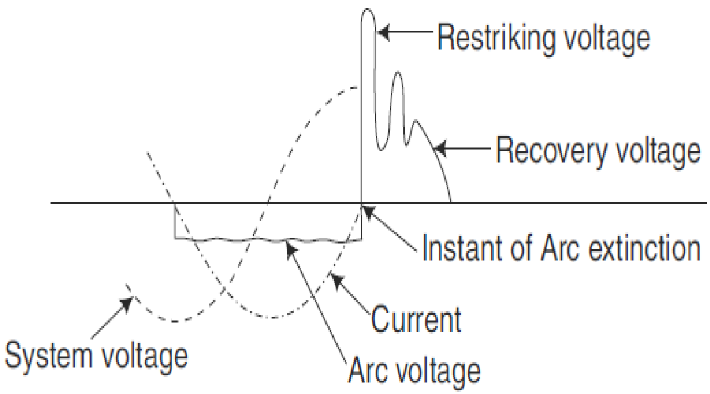


Fig. 14.7 *Restriking and recovery voltage*

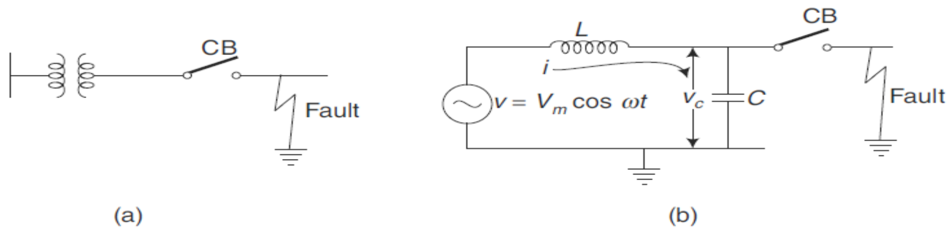


Fig. 14.8 *(a) Fault on a feeder near circuit breaker (b) Equivalent electrical circuit for analysis of restriking voltage*

The mathematical expression for the transient condition is as follows:

$$L\frac{di}{dt} + \frac{1}{c}\int i\,dt = V_m \cos \omega t$$

Substituting $V_m \cos \omega t \approx V_m$, the Eq. (14.3) can be written as

$$L\frac{di}{dt} + \frac{1}{c}\int i\,dt = V_m$$

$$i = \frac{dq}{dt} = \frac{d(cv_c)}{dt}$$

where, v_c = voltage across the capacitor = Restriking voltage

Therefore, $$\frac{di}{dt} = \frac{d^2(cv_c)}{dt^2} = c\frac{d^2v_c}{dt^2}$$

$$\frac{1}{c}\int i\,dt = \frac{q}{c} = v_c$$

Substituting these values in Eq. (14.4), we get

$$LC\frac{d^2v_c}{dt^2} + v_c = V_m$$

Taking Laplace Transform of both sides of Eq. (14.8), we get

$$LCS^2v_c(s) + v_c(s) = \frac{V_m}{s}$$

where, $v_c(s)$ is the Laplace Transform of v_c.

Other terms are zero as initially $q = 0$ at $t = 0$

or $$v_c(s)\,[LCS^2 + 1] = \frac{V_m}{s}$$

or $$v_c(s) = \frac{V_m}{s(LCS^2 + 1)} = \frac{V_m}{LCS\left(s^2 + \frac{1}{LC}\right)}$$

$$w_n = \frac{1}{\sqrt{LC}}, \quad \text{therefore, } \frac{1}{LC} = w_n^2$$

or $$v_c(s) = \frac{w_n^2 V_m}{s(s^2 + w_n^2)} = \frac{w_n V_m}{s}\left(\frac{w_n}{s^2 + w_n^2}\right)$$

Taking the inverse Laplace of Eq. (14.9), we get

$$v_c(t) = w_n V_m \int_0^t \sin \omega_n t$$

$$= w_n V_m \left[\frac{-\cos \omega_n t}{w_n}\right]_0^t$$

As $v_c(t) = 0$ at $t = 0$, constant $= 0$.

or $$v_c(t) = V_m\,(1 - \cos \omega_n t)$$

This is the expression for the restriking voltage.

The Rate of Rise of Restriking Voltage (RRRV)

$$= \frac{d}{dt}\,[V_m\,(1 - \cos \omega_n t)]$$

r $$\text{RRRV} = V_m w_n \sin \omega_n t$$

The maximum value of RRRV occurs when $\omega_n t = \pi/2$ i.e., when $t = \pi/2\omega_n$,

Hence, the maximum value of RRRV $= V_m \omega_n$

PROBLEMS OF CIRCUIT INTERRUPTION:

Current chopping:

It is the phenomenon of current interruption before the natural current zero is reached. Current chopping mainly occurs in air-blast circuit breakers because they retain the same extinguishing power irrespective of the magnitude of the current to be interrupted. When breaking low currents (e.g., transformer magnetizing current) with such breakers, the powerful de-ionizing effect of air-blast causes the current to fall abruptly to zero well before the natural current zero is reached. This phenomenon is known as current chopping and results in the production of high voltage transient across the contacts of the circuit breaker as discussed below:

Suppose the arc current is i when it is chopped down to zero value as shown by point a in Fig. (ii). As the chop occurs at current i, therefore, the energy stored in inductance is L i^2 /2. This energy will be transferred to the capacitance C, charging the latter to a prospective voltage e

$$\frac{1}{2} L\, i^2 = \frac{C\, e^2}{2}$$

$$e = i\sqrt{\frac{L}{C}} \text{ volts}$$

The prospective voltage e is very high as compared to the dielectric strength gained by the gap so that the breaker restrike. As the de-ionizing force is still in action, therefore, chop occurs again but the arc current this time is smaller than the previous case. This induces a lower prospective voltage to re-ignite the arc. In fact, several chops may occur until a low enough current is interrupted which produces insufficient induced voltage to re-strike across the breaker gap. Consequently, the final interruption of current takes place.

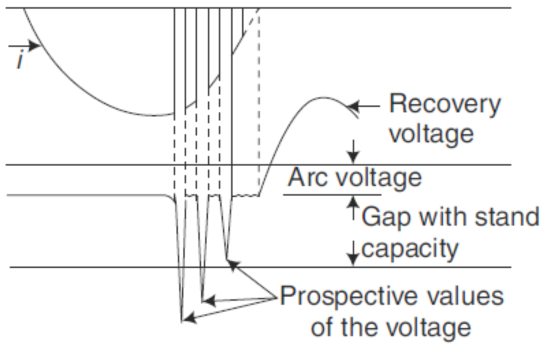


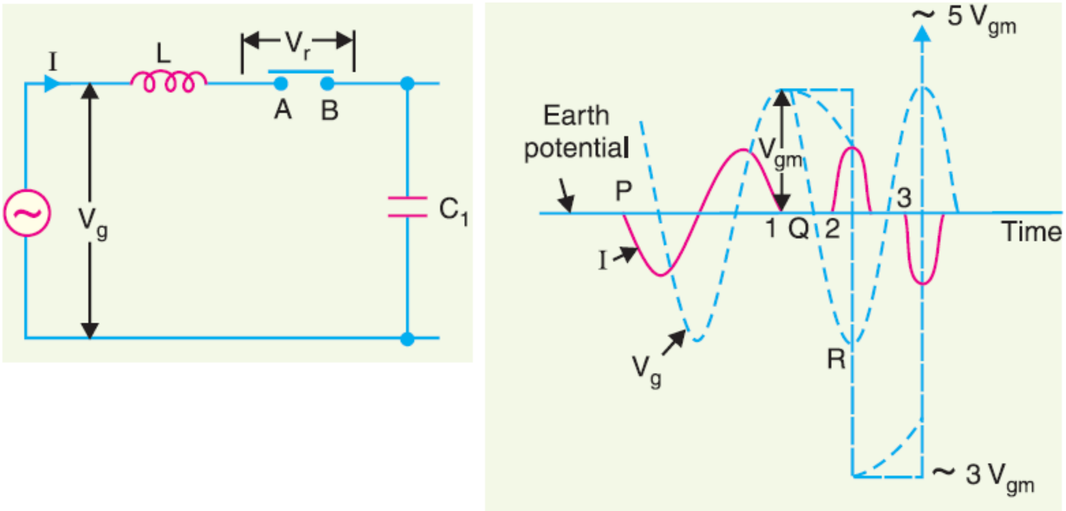
Fig. 14.12 *Current chopping*

Excessive voltage surges due to current chopping are prevented by shunting the contacts of the breaker with a resistor (resistance switching) such that re ignition is unlikely to occur.

Capacitive current breaking or interruption of capacitive currents.

Another cause of excessive voltage surges in the circuit breakers is the interruption of capacitive currents. Examples of such instances are opening of an unloaded long transmission line, disconnecting a capacitor bank used for power factor improvement etc. Consider the simple equivalent circuit of an unloaded transmission line shown in Fig.19.20. Such a line, although unloaded in the normal sense, will actually carry a capacitive current I on account of appreciable amount of capacitance C between the line and the earth.

Let us suppose that the line is opened by the circuit breaker at the instant when line capacitive current is zero [point 1 in Fig. At this instant, the generator voltage Vg will be maximum (i.e. V gm) lagging behind the current by 90º. The opening of the line leaves a standing charge on it (i.e., end B of the line) and the capacitor C1 is charged to Vgm. However, the generator end of the line (i.e., end A of the line) continues its normal sinusoidal variations. The voltage Vr across the circuit breaker will be the difference between the voltages on the respective sides. Its initial value is zero (point 1) and increases slowly in the beginning. But half a cycle later [point R in Fig.] the potential of the circuit breaker contact "A" becomes maximum negative which causes the voltage across the breaker (V r) to become 2 V gm. This voltage may be sufficient to restrike the arc. The two previously separated parts of the circuit will now be joined by an arc of very low resistance. The line capacitance discharges at once to reduce the voltage across the circuit breaker, thus setting up high frequency transient. The peak value of the initial transient will be twice the voltage at that instant i.e., −4 V gm. This will cause the transmission voltage to swing to −4V gm to + V gm i.e., −3V gm.



The re-strike arc current quickly reaches its first zero as it varies at natural frequency. The voltage on the line is now −3 Vgm and once again the two halves of the circuit are separated and the line is isolated at this potential. After about half a cycle further, the aforesaid events are repeated even on more formidable scale and the line may be left with a potential of 5V gm above earth potential. Theoretically, this phenomenon may proceedinfinitely increasing the voltage by successive increment of 2 times V gm.

While the above description relates to the worst possible conditions, it is obvious that if the gap breakdown strength does not increase rapidly enough, successive re-strikes can build up a dangerous voltage in the open circuit line. However, due to leakage and corona loss, the maximum voltage on the line in such cases is limited to 5 V gm.

Resistance Switching:

It has been discussed above that current chopping, capacitive current breaking etc. give rise to severe voltage oscillations. These excessive voltage surges during circuit interruption can be prevented by the use of shunt resistance R connected across the circuit breaker contacts as shown in the equivalent circuit in Fig. 19.22. This is known as resistance switching.

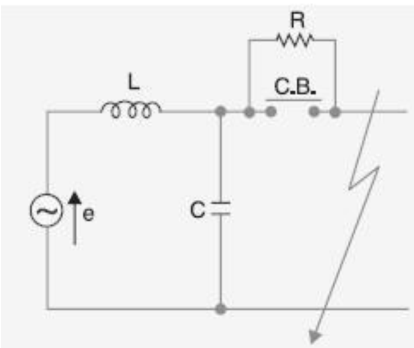


Fig. 19.22

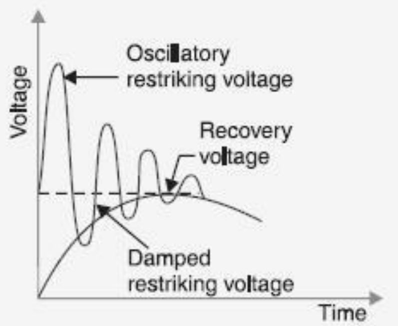


Fig. 19.23

Referring to Fig. 19.22, when a fault occurs, the contacts of the circuit breaker are opened and an arc is struck between the contacts. Since the contacts are shunted by resistance R, a part of arc current flows through this resistance. This results in the decrease of arc current and an increase in the rate of de-ionization of the arc path. Consequently, the arc resistance is increased. The increased arc resistance leads to a further increase in current through shunt resistance. This process continues until the arc current becomes so small that it fails to maintain the arc. Now, the arc is extinguished and circuit current is interrupted.

Circuit Breaker Ratings:

A circuit breaker may be called upon to operate under all conditions. However, major duties are imposed on the circuit breaker when there is a fault on the system in which it is connected. Under fault conditions, a circuit breaker is required to perform the following three duties:

(i) It must be capable of opening the faulty circuit and breaking the fault current.

(ii) It must be capable of being closed on to a fault.

(iii) It must be capable of carrying fault current for a short time while another circuit breaker (in series) is clearing the fault.

Corresponding to the above mentioned duties, the circuit breakers have three ratings viz.

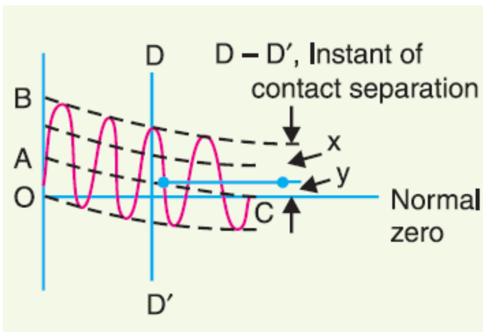
1. Breaking capacity

2. Making capacity and

3. Short-time capacity.

Breaking capacity: **It is current (r.m.s.) that a circuit breaker is capable of breaking at given recovery voltage and under specified conditions (e.g., power factor, rate of rise of restriking voltage).**

The breaking capacity is always stated at the r.m.s. value of fault current at the instant of contact separation. When a fault occurs, there is considerable asymmetry in the fault current due to the presence of a d.c. component. The d.c. component dies away rapidly, a typical decrement factor being 0·8 per cycle. Referring toFig., the contacts are separated at DD´ At this instant, the fault current has



x = maximum value of a.c. component

y = d.c. component

Symmetrical breaking current = r.m.s. value of a.c. component

Asymmetrical breaking current = r.m.s. value of total current

It is a common practice to express the breaking capacity in MVA by taking into account the rated breaking current and rated service voltage. Thus, if I is the rated breaking current in amperes and V is the rated service line voltage in volts, then for a 3-phase circuit,

Breaking capacity = $\sqrt{3}$*V*I MVA

Making capacity:

There is always a possibility of closing or making the circuit under short circuit conditions. The capacity of a breaker to make current depends upon its ability to withstand and close successfully against the effects of electromagnetic forces. These forces are proportional to the square of maximum instantaneous current on closing. Therefore, making capacity is stated in terms of a peak value of current instead of r.m.s. value.

The peak value of current (including d.c. component) during the first cycle of current wave after the closure of circuit breaker is known as making capacity.

Short-time rating:

It is the period for which the circuit breaker is able to carry fault current while remaining closed. Sometimes a fault on the system is of very temporary nature and persists for 1 or 2 seconds after which the fault is automatically cleared. In the interest of continuity of supply, the breaker should not trip in such situations. This means that circuit breakers should be able to carry high current safely for some specified period while remaining closed i.e., they should have proven short-time rating. However, if the fault persists for duration longer than the specified time limit, the circuit breaker will trip, disconnecting the faulty section.

The short-time rating of a circuit breaker depends upon its ability to withstand the electromagnetic force effects and the temperature rise.

Normal current rating:

It is the r.m.s. value of current which the circuit breaker is capable of carrying continuously at its rated frequency under specified conditions. The only limitation in this case is the temperature rise of current-carrying parts.

Classification of Circuit Breakers:

There are several ways of classifying the circuit breakers. However, the most general way of classification is on the basis of medium used for arc extinction. The medium used for arc extinction is usually oil, air, sulphur hexafluoride (SF6) or vacuum. Accordingly, circuit breakers may be classified into:

1. **Oil circuit breakers:** which employ some insulating oil (e.g., transformer oil) for arc extinction?

2. **Air-blast circuit breakers:** in which high pressure air-blast is used for extinguishing the arc.

3. **Sulphur hexafluoride circuit breakers:** in which sulphur hexafluoride (SF6) gas is used for arc extinction.

4. **Vacuum circuit breakers:** in which vacuum is used for arc extinction.

Sulphur Hexafluoride (SF6) Circuit Breakers:

In such circuit breakers, sulphur hexafluoride (SF6) gas is used as the arc quenching medium. The SF6 is an electro-negative gas and has a strong tendency to absorb free electrons. The contacts of the breaker are opened in a high pressure flow of SF6 gas and an arc is struck between them. The conducting free electrons in the arc are rapidly captured by the gas to form relatively immobile negative ions. This loss of conducting electrons in the arc quickly builds up enough insulation strength to extinguish the arc. The SF6 circuit breakers have been found to be very effective for high power and high voltage service.

Construction:

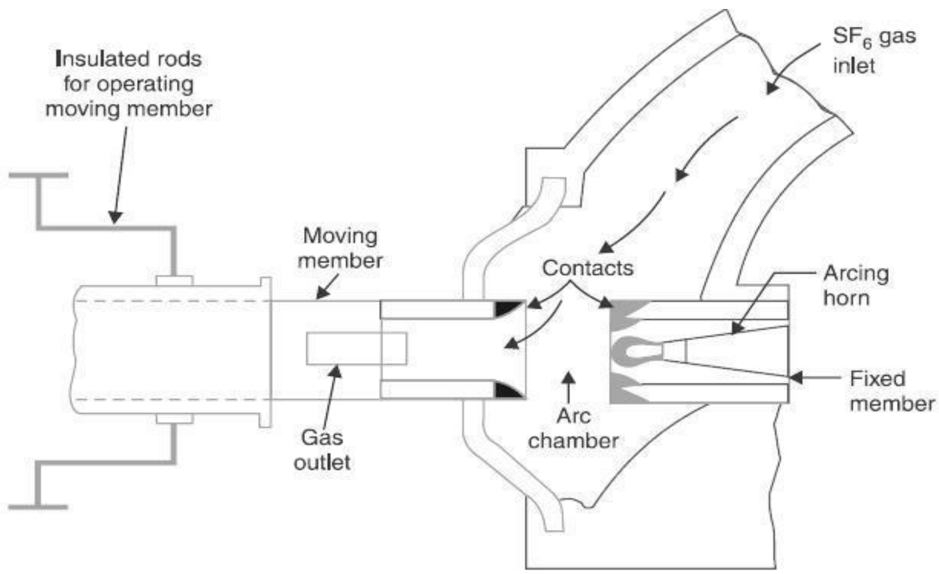


Fig. 19.11

Fig. 19.11 shows the parts of a typical SF6 circuit breaker. It consists of fixed and moving contacts enclosed in a chamber (called arc interruption chamber) containing SF6 gas. This chamber is connected to SF6 gas reservoir. When the contacts of breaker are opened, the valve mechanism permits a high pressure SF6 gas from the reservoir to flow towards the arc interruption chamber. The fixed contact is a hollow cylindrical current carrying contact fitted with an arc horn. The moving contact is also a hollow cylinder with rectangular holes in the sides to permit the SF6 gas to let out through these holes after flowing along and across the arc. The tips of fixed contact, moving contact and arcing horn are coated with copper-tungsten arc resistant material. Since SF6 gas is Costly, it is reconditioned and reclaimed by suitable auxiliary system after each operation of the breaker.

Working:

In the closed position of the breaker, the contacts remain surrounded by SF6 gas at a pressure of about 2·8kg/cm. When the breaker operates, the moving contact is pulled apart and an arc is struck between the contacts. The movement of the moving contact is synchronized with the opening of a valve which permits SF6 gas at 14 kg/cm pressure from the reservoir to the arc interruption chamber. The high pressure flow of SF6 rapidly absorbs the free electrons in the arc path to form immobile negative ions which are ineffective as charge carriers. The result is that the medium between the contacts quickly builds up high dielectric strength and causes the extinction of the arc. After the breaker operation (i.e., after arc extinction), the valve is closed by the action of a set of springs.

Advantages:

Due to the superior arc quenching properties of SF6 gas, the SF6 circuit breakers have many advantages over oil or air circuit breakers. Some of them are listed below:

1. Due to the superior arc quenching property of SF6, such circuit breakers have very short arcing time.

2. Since the dielectric strength of SF6 gas is 2 to 3 times that of air, such breakers can interrupt much larger currents.

3. The SF6 circuit breaker gives noiseless operation due to its closed gas circuit and no exhaust to atmosphere unlike the air blast circuit breaker

4. The closed gas enclosure keeps the interior dry so that there is no moisture problem.

5. There is no risk of fire in such breakers because SF6 gas is non-inflammable.

6. There are no carbon deposits so that tracking and insulation problems are eliminated.

7. The SF6 breakers have low maintenance cost, light foundation requirements and minimum auxiliary equipment.

8. Since SF6 breakers are totally enclosed and sealed from atmosphere, they are particularly suitable where explosion hazard exists e.g., coal mines.

Disadvantages:

1. SF6 breakers are costly due to the high cost of SF6.

2. Since SF6 gas has to be reconditioned after every operation of the breaker, additional equipment is required for this purpose.

Applications:

A typical SF6 circuit breaker consists of interrupter units each capable of dealing with currents up to 60 kA and voltages in the range of 50—80 kV. A number of units are connected in series according to the systemvoltage. SF6 circuit breakers have been developed for voltages 115 kV to 230 kV, power ratings 10 MVA to 20 MVA and interrupting time less than 3 cycles.

Vacuum Circuit Breakers (VC B):

In such breakers, vacuum (degree of vacuum being in the range from 10 to 10 torr) is used as the arc quenching medium. Since vacuum offers the highest insulating strength, it has far superior arc quenching properties than any other medium. For example, when contacts of a breaker are opened in vacuum, the interruption occurs at first current zero with dielectric strength between the contacts building up at a rate thousands of times higher than that obtained with other circuit breakers.

Principle: The production of arc in a vacuum circuit breaker and its extinction can be explained as follows: When the contacts of the breaker are opened in vacuum (10 to 10 torr), an arc is produced between the contacts by the ionization of metal vapours of contacts. However, the arc is quickly extinguished because the metallic vapours, electrons and ions produced during arc rapidly condense on the surfaces of the circuit breaker contacts, resulting in quick recovery of dielectric strength. The reader may note the salient feature of vacuum as an arc quenching medium. As soon as the arc is produced in vacuum, it is quickly extinguished due to the fast rate of recovery of dielectric strength in vacuum.

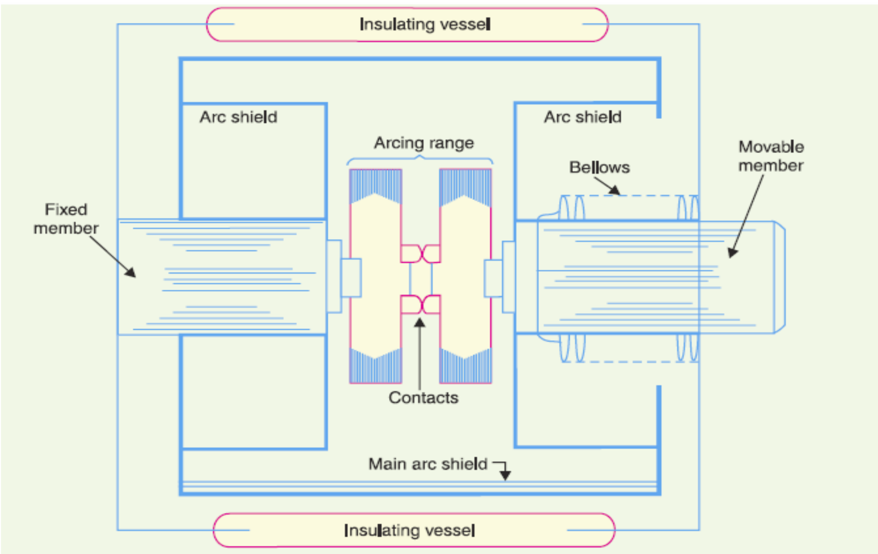


Fig. 19.12

Construction:

Fig. 19.12 shows the parts of a typical vacuum circuit breaker. It consists of fixed contact, moving contact and arc shield mounted inside a vacuum chamber. The movable member is connected to the control mechanism by stainless steel bellows. This enables the permanent sealing of the vacuum chamber so as to eliminate the possibility of leak. A glass vessel or ceramic vessel is used as the outer insulating body. The arc shield prevents the deterioration of the internal dielectric strength by preventing metallic vapours falling on the inside surface of the outer insulating cover.

Working:

When the breaker operates, the moving contact separates from the fixed contact and an arc is struck between the contacts. The production of arc is due to the ionization of metal ions and depends very much upon the material of contacts. The arc is quickly extinguished because the metallic vapours, electrons and ions produced during arc are diffused in a short time and seized by the surfaces of moving and fixed members and shields. Since vacuum has very fast rate of recovery of dielectric strength, the arc extinction in a vacuum breaker occurs with a short contact separation (say 0·625 cm).

Vacuum circuit breakers have the following advantages:

1. They are compact, reliable and have longer life.

2. There is no generation of gas during and after operation.

3. They can interrupt any fault current. The outstanding feature of a V C B is that it can break any heavy fault current perfectly just before the contacts reach the definite open position.

4. They require little maintenance and are quiet in operation.

5. They have low arc energy.

6. They have low inertia and hence require smaller power for control mechanism.

Applications:

For a country like India, where distances are quite large and accessibility to remote areas difficult, the installation of such outdoor, maintenance free circuit breakers should prove a definite advantage. Vacuum circuit breakers are being employed for outdoor applications ranging from 22 kV to 66 kV. Even with limited rating of say 60 to 100 MVA, they are suitable for a majority of applications in rural areas.

GAS INSULATED SUBSTATION OR SWITCHGEAR (GIS)

Conventional air-insulated transmission and distribution substations are of substantial size due to the poor dielectric strength of air. They also suffer variations in the dielectric capability of air to withstand varying ambient conditions and deterioration of the exposed components due to corrosive and oxidising nature of the environment.
In order to enhance the life and reliability of power transmission and distribution substations, it is desirable to protect the substation components from corrosion and oxidisation due to

environment. The use of sulphur hexafluoride (SF6) gas with higher dielectric strength instead of air helps in manifold reduction in the size
of the substation components. On the other hand, the grounded metal-enclosure makes the substation equipment safe, as the live components are no longer within the reach of the operator. As the metal enclosure of the equipment is solidly grounded, the electric field intensity, at the enclosure surface, is reduced
to zero. Using this design philosophy, various substation equipment such as circuit breakers, busbars, earthing switches, isolators, instrument transformers (current and voltage transformers), etc., are housed in separate metal enclosed modules field with pressurised SF6 gas. The assembly of such equipment at a substation is defined Gas Insulated Substation (GIS). The term GIS is also sometimes used to refer to Gas Insulated Switchgear.

Modules/Components of GIS
The following are the principal modules insulated by SF6 gas in a GIS:
(i) Busbar
(ii) Isolator or disconnector
(iii) Circuit breaker
(iv) Current transformer
(v) Earthing switch

The auxiliary gas insulated modules/accessories (excluding control panel) required
to complete a GIS are
(i) Instrument voltage transformer
(ii) Surge and lightning arrester
(iii)Terminations

Busbar
The busbar which is one of the most elementary components of the GIS system, is of different lengths to cater to the requirement of circuit. Co-axial busbars are commonly used in isolated-phase GIS as this configuration results in an optimal stress distribution.

Isolator or Disconnector
Isolators are placed in series with the circuit breaker to provide additional protection and physical isolation. In a circuit, two isolators are generally used, one on the line side and the other on the feeder side.

Circuit Breaker
The circuit breaker is the most critical module of a GIS system. It is metal-clad and utilises SF6 gas, both for insulation and fault interruption. Puffer type SF6 circuit breakers are commonly used to accomplish fault current interruption in GIS.

Earthing Switch
Earthing switches used for GIS system are of two types, namely, maintenance earthing switch and fast earthing switch. The maintenance earthing switch is a slow device used to ground the high voltage conductors during maintenance schedule, in order to ensure the safety of the maintenance staff. On the other hand, the fast earthing switch is used to protect the circuit-connected instrument voltage transformer from core saturation caused by direct current flowing

through its primary as a result of remnant charge (stored online during isolation/switching off the line).

Classification of GIS

Gas-insulated substations (GIS) are classified according to the type of modules or the configuration. The following configurations of modules are generally used in GIS.

(i) Isolated-phase (segregated-phase) module.
(ii) Three-phase common modules
(iii) Hybrid modules
(iv) Compact modules
(v) Highly integrated systems

The isolated-phase GIS module contains an assembly of individual circuit elements such as a pole of circuit breaker, a single pole isolator, one-phase assembly of a current transformer, etc. A single-phase circuit is formed by using individual components and pressurising the element with SF6 gas forming a gas-tight circuit. A complete three-phase GIS section/bay is formed by three such circuits, arranged side by side. Hybrid systems use a suitable combination of isolated-phase and three-phase common elements. Compact GIS system are essentially three-phase common systems, with more than one functional elements in one enclosure.

Highly integrated systems (HIS) are single unit metal enclosed and SF6 gas insulated substations which are gaining user appreciation as this equipment provides a total substation solution for outdoor substations.

Advantages of GIS

(i) Compactness GIS are compact in size and the space occupied by them is about 15% of that of a conventional air insulated outdoor substation. The compact size of GIS offers a practical solution to vertically upgrade the existing substation and to meet the ever-increasing power demand in developing countries.

(ii) Cost-effective, reliable and maintenance-free GIS offer a cost-effective, reliable and maintenance free alternative to conventional air insulated substations.

(iii) Protection from pollution The moisture, pollution, dust etc. have little influence on GIS.

(iv) Reduced installation time The modular construction reduces the installation time to a few weeks.

(v) Superior arc interruption SF6 gas used in the circuit-breaker has superior arc quenching property.

(vi) Reduced switching over voltages The over voltages while closing and opening line, cables, motors, capacitors etc. are low.

(vii) Increased safety As the metal enclosures of the various modules are at ground potential, there is no possibility of accidental contact by service personnel to live parts.

Disadvantages of GIS

(i) GIS has high cost as compared to conventional outdoor substation.

(ii) Requirement of cleanliness in GIS are very stringent as dust or moisture can cause internal flashover.

(iii) Since GIS are generally indoor, they need separate building which is generally not required for conventional outdoor substations.

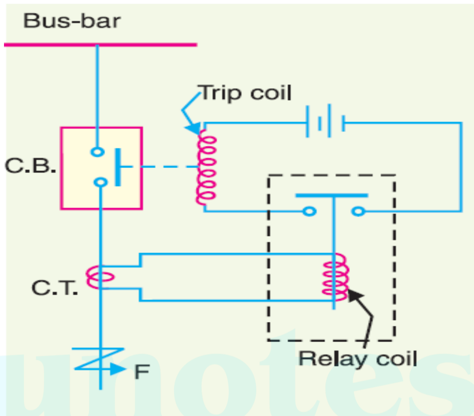
(iv) GIS can have excessive damage in case of internal fault and there may be long outage periods as repair of damaged part may be difficult at site.

(v) Maintenance of adequate stock of SF6 gas is essential. Procurement of gas and supply of gas is a problem.

Protective Relays

A **protective relay** *is a device that detects the fault and initiates the operation of the circuit breaker to ioslate the defective element from the rest of the system.*

The relays detect the abnormal conditions in the electrical circuits by constantly measuring the electrical quantities which are different under normal and fault conditions. The electrical quantities which may change under fault conditions are voltage, current, frequency and phase angle. Through the changes in one or more of these quantities, the faults signal their presence, type and location to the protective relays. Having detected the fault, the relay operates to close the trip circuit of the breaker. This results in the opening of the breaker and disconnection of the faulty circuit.



i) First part is the primary winding of a current transformer (C.T.) which is connected in series with the line to be protected.
(*ii*) Second part consists of secondary winding of C.T. and the relay operating coil.
(*iii*) Third part is the tripping circuit which may be either a.c. or d.c. It consists of a source of supply, the trip coil of the circuit breaker and the relay stationary contacts.

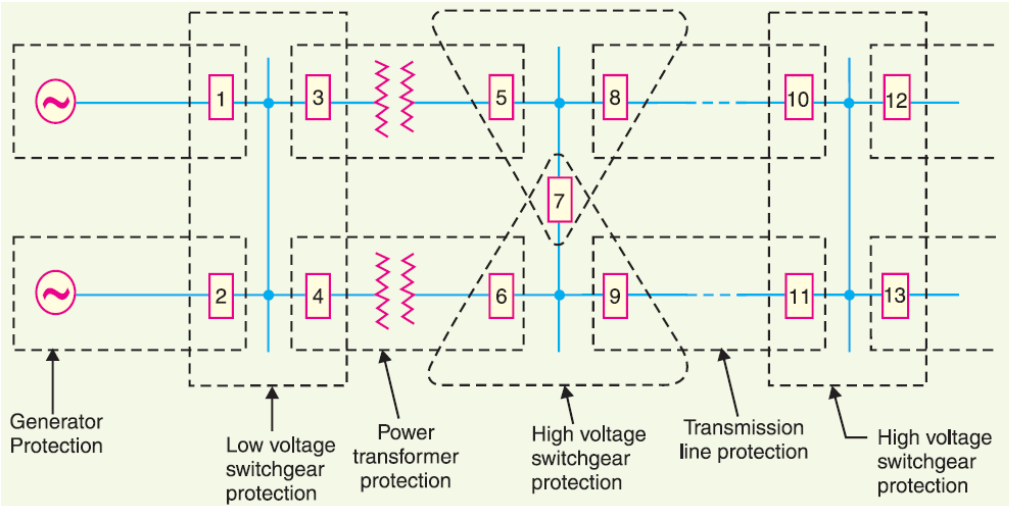
Fundamental Requirements of Protective Relaying

(i) Selectivity (*ii*) speed (*iii*) sensitivity(*iv*) reliability (*v*) simplicity (*vi*) economy

(*i*) Selectivity. *It is the ability of the protective system to select correctly that part of the systemin trouble and disconnect the faulty part without disturbing the rest of the system.*
(*ii*) Speed. The relay system should disconnect the faulty section as fast as possible
(*iii*) Sensitivity. *It is the ability of the relay system to operate with low value of actuating quantity.*
(*iv*) Reliability. It is the ability of the relay system to operate under the pre-determined conditions. Without reliability, the protection would be rendered largely ineffective and could even become a liability.
(*v*) Simplicity. The relaying system should be simple so that it can be easily maintained. Reliability is closely related to simplicity. The simpler the protection scheme, the greater will be its reliability.

(*vi*) **Economy.** The most important factor in the choice of a particular protection scheme is the economic aspect. Sometimes it is economically unjustified to use an ideal scheme of protection and a compromise method has to be adopted.

ZONES OF PROTECTION



In order to provide selectivity to the system, it is a usual practice to divide the entire system into several protection zones. When a fault occurs in a given zone, then only the circuit breakers within that zone will be opened. This will isolate only the faulty circuit or apparatus, leaving the healthy circuits intact. The system can be divided into the following protection zones:

(*a*) generators (*b*) low-tension switchgear (*c*) transformers (*d*) high-tension switchgear (*e*) transmission lines

CLASSIFICATION OF RELAYS

1. ELECTROMECHANICAL They work on the following two main operating principles:
(*i*) Electromagnetic attraction
(*ii*) Electromagnetic induction
2. STATIC
3. MICROPROCESSOR BASED

Electromagnetic Attraction Relays
Electromagnetic attraction relays operate by virtue of an armature being attracted to the poles of an electro magnet or a plunger being drawn into a solenoid. Such relays may be actuated by d.c. or a.c. quantities.

Attracted armature type relay.

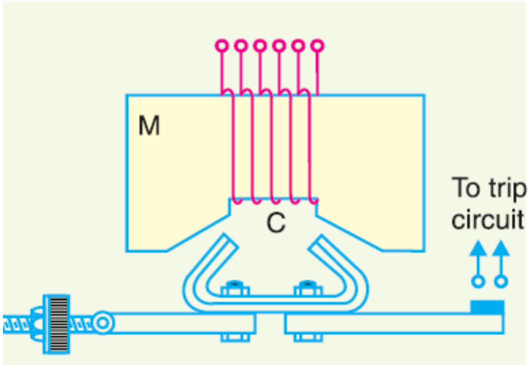


Fig. shows the schematic arrangement of an attracted armature type relay. It consists of a laminated electromagnet *M* carrying a coil *C* and a pivoted laminated armature. The armature is balanced by a counterweight and carries a pair of spring contact fingers at its free end. Under normal operating conditions, the current through the relay coil *C* is such that counterweight holds the armature in the position shown. However, when a short-circuit occurs, the current through the relay coil increases sufficiently and the relay armature is attracted upwards. The contacts on the relay armature bridge a pair of stationary contacts attached to the relay frame. This completes the trip circuit which results in the opening of the circuit breaker and, therefore, in the disconnection of the faulty circuit. The minimum current at which the relay armature is attracted to close the trip circuit is called *pickup* current.

Induction Relays

Electromagnetic induction relays operate on the principle of induction motor and are widely used for protective relaying purposes involving a.c. quantities. They are not used with d.c. quantities owing to the principle of operation. An induction relay essentially consists of a pivoted aluminium disc placed in two alternating magnetic fields of the same frequency but displaced in time and space. The torque is produced in the disc by the interaction of one of the magnetic fields with the currents induced in the disc by the other.

The following three types of structures are commonly used for obtaining the phase difference in the fluxes and hence the operating torque in induction relays:

(*i*) shaded-pole structure

(*ii*) watthour-meter or double winding structure

(*iii*) induction cup structure

(*i*) Shaded-pole structure. The general arrangement of shaded-pole structure is shown in Fig. It consists of a pivoted aluminium disc free to rotate in the air-gap of an electromagnet. One half of each pole of the magnet is surrounded by a copper band known as *shading ring.* The alternating flux Φ_s in the shaded portion of the poles will, owing to the reaction of the current induced in the ring, lag behind the flux Φ_u in the unshaded portion by an angle α. These two a.c. fluxes differing in phase will produce the necessary torque to rotate the disc. As proved earlier, the driving torque T is

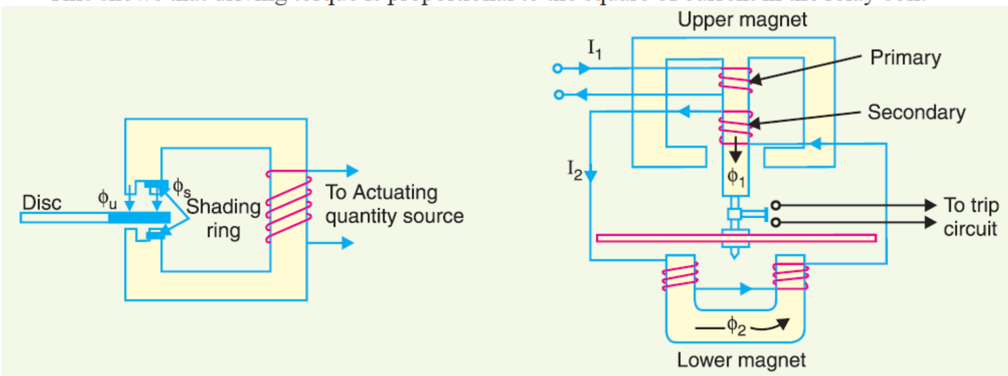
given by;

$$T \propto \phi_s \phi_u \sin \alpha$$

Assuming the fluxes ϕ_s and ϕ_u to be proportional to the current I in the relay coil,

$$T \propto I^2 \sin \alpha$$

This shows that driving torque is proportional to the square of current in the relay coil.

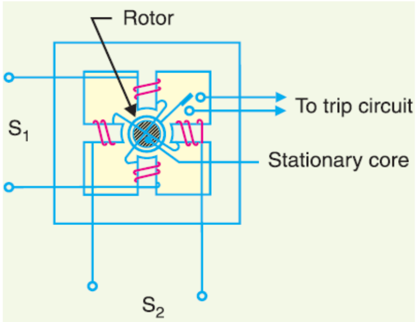


(*ii*) Watthour-meter structure. This structure gets its name from the fact that it is used in watthour meters. The general arrangement of this type of relay is shown in Fig. It consists of a pivoted aluminium disc arranged to rotate freely between the poles of two electromagnets. The upper electromagnet carries two windings; the primary and the secondary. The primary winding carries the relay current $I1$ while the secondary winding is connected to the winding of the lower magnet. The primary current induces e.m.f. in the secondary and so circulates a current $I2$ in it. The flux Φ2 induced in the lower magnet by the current in the secondary winding of the upper magnet will lag behind Φ1 by an angle α. The two fluxes Φ1and Φ2 differing in phase by α will produce a driving torque on the disc proportional to Φ1Φ2 sin α.

An important feature of this type of relay is that its operation can be controlled by opening or closing the secondary winding circuit. If this circuit is opened, no flux can be set by the lower magnet however great the value of current in the primary winding may be and consequently no torque will be produced. Therefore, the relay can be made inoperative by opening its secondary winding circuit.

(*iii*) Induction cup structure. Fig. shows the general arrangement of an induction cup structure. It most closely resembles an induction motor, except that the rotor iron is stationary, only the rotor conductor portion being free to rotate. The moving element isa hollow cylindrical rotor which turns on its axis. The rotating field

is produced by two pairs of coils wound on four poles as shown. The rotating field induces currents in the cup to provide the necessary driving torque. If Φ1 and Φ2 represent the fluxes produced by the respective pairs of poles, then torque produced is proportional to Φ1Φ2 sin α, where α is the phase difference between the two fluxes.



Thermal Relays

These relays utilise the electro-thermal effect of the actuating current for their operation They are widely used for the protection of small motors against overloading and unbalanced currents. The thermal element is a bimetallic strip, usually wound into a spiral to obtain a greater length, resulting in a greater sensitivity.

A bimetallic element consists of two metal strips of different coefficients of thermal expansion, joined together. When it heats up one strip expand more than other. This results in the bending of the bimetallic strip. The thermal element can be heated directly by passing the actuating current through the strip, but usually a heater coil is employed. When the bimetallic element heats up, it bends and deflects, thereby closing the relay contacts.

When the strip is in a spiral form, the unequal expansions of the two metals causes the unwinding of the spiral, which results in the closure of the contacts.

Fig. (a) shows a simple arrangement to indicate the operating principle. Fig. (b) shows a spiral form. Unmetallic strips are also used as thermal elements in a hair-pin like shape, as shown in fig.(c) When the strip gets heated, it expands and closes the contacts.

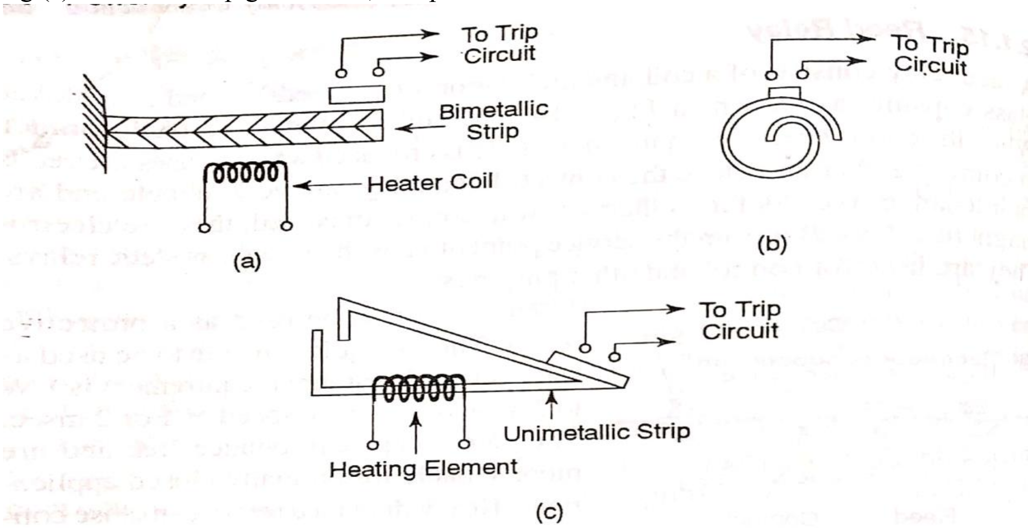
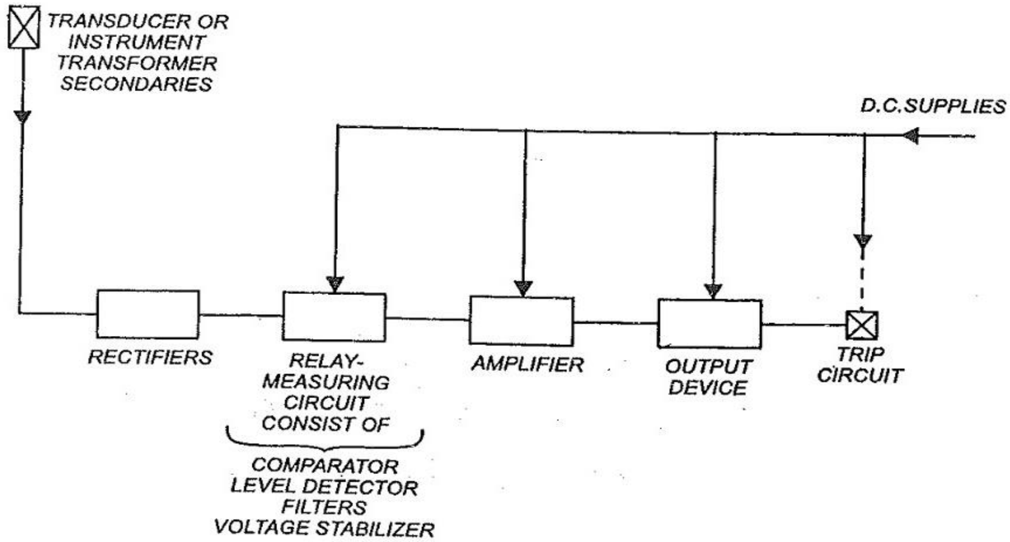


FIGURE 2.15 (a) Bimetallic thermal relay (b) Bimetallic spiral type thermal relay (c) Unimetallic thermal relay

STATIC RELAYS OR SOLIDSTATE RELAYS



Block diagram of a static relay-simplified.

Static relay components

Diodes, zeners, transistors, thyristors, rectifiers, voltage rectified circuits, smoothing circuits, transistor amplifier, filter circuits, logic circuits, multivibrators, differentiating circuits, integrating circuits, operational amplifier, level detector, time delay circuits, output circuits, DC auxiliary supply, surge absorbers, /surge suppressors, comparators.

Here the relaying quantity, ie, the output of a CT or P.T. or a transducer is rectified by a rectifier. Static relays can be arranged to respond electrical inputs. The other types of inputs such as heat, light, magnetic field, traveling waves etc, can suitably be converted into equivalent analogue or digital signals and then supplied to the static relay. The rectified output is fed to a measuring unit comprising comparators, level detectors, filters, logic circuits, voltage stabilizers etc. The output is actuated when the dynamic input (the relaying quantity) attains a threshold value. This output of the measuring unit is amplified by amplifier and fed to the output unit device, which is usually an electromagnetic one. The output unit energizes the trip coil of the circuit breaker when the relay operates.

Merits and Demerits of Static Relays:

The advantages of static relays over electromagnetic relays are as follows:

- Low burden on C.T.s and P.T.s. – The static relays consume less power and in most of these cases they draw power from the auxiliary dc supply.
- Fast response
- Long life
- High resistance to shock and vibration
- Less maintenance due to the absence of moving parts and bearings.
- Frequent operations cause no deterioration

- Quick resetting and absence of overshoot.
- Compact size.
- Greater sensitivity as amplification can be provided easily.
- Complex relaying Characteristics can easily be obtained.
- Logic circuits can be used for complex protective schemes.

The demerits of static relays are as follows:

Static relays are temperature sensitive. Their characteristics may vary with the variation of temperature. Temperature compensation can be made by using thermistors and by using digital techniques of measurements, etc.
Static relays are sensitive to voltage transients. The semiconductor components may get damaged due to voltage spikes. Filters and shielding can be used for their protection against voltage spikes.
Static relays need an auxiliary power supply. This can however be easily supplied by a battery or a stabilized power supply

Comparators

When faults occur on a system, the magnitude of voltage and current, and phase angle between voltage and current may change. These quantities during faulty conditions are different from those under healthy conditions. The static relay circuitry is designed to recognise the changes and to distinguish between healthy and faulty conditions. Either magnitudes of voltage/current (or corresponding derived quantities) are compared or phase angle between voltage and current (or corresponding derived quantities) are measured by the static relay circuitry and a trip signal is sent to the circuit breaker when a fault occurs.
The part of the circuitry which compares the two actuating quantities either in amplitude or phase is known as the comparator.

There are two types of comparators: Amplitude comparator and Phase comparator

Amplitude Comparator
An amplitude comparator compares the magnitude of two input quantities irrespective of the angle between them. One of the inputs is the operating quantity and the other a restraining quantity. When the amplitude of the operating quantity exceeds that of restraining quantity, the relay sends a tripping signal to the circuit breaker.

Phase Comparator
A phase comparator compares two input quantities in phase angle (vertically) irrespective of the magnitude and operates if the phase angle between them is $< =90^{o}$

2.3.3 Duality between Amplitude and Phase Comparators

An amplitude comparator can be converted to a phase comparator and vice-versa if the input quantities to the comparator are modified. The modified input quantities are the sum and difference of the original two input quantities. To understand this fact, consider the operation of an amplitude comparator which has two input signals M and N as shown in Fig. 2.16 (a). It operates when $|M| > |N|$. Now change the input quantities to $(M + N)$ and $(M - N)$ as shown in Fig. 2.16(b). As its circuit is designed for amplitude comparison, now with the changed input, it will operate when $|M + N| > |M - N|$. This condition will be satisfied only when the phase angle between M and N is less than 90°. This has been illustrated with the vector diagram shown in Fig. 2.17. It means that the comparator with the modified inputs has now become a phase comparator for the original input signals M and N.

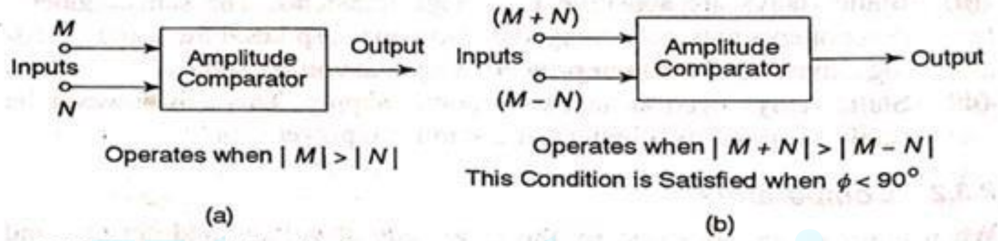


FIGURE 2.16 (a) Amplitude comparator (b) Amplitude comparator used for phase comparison

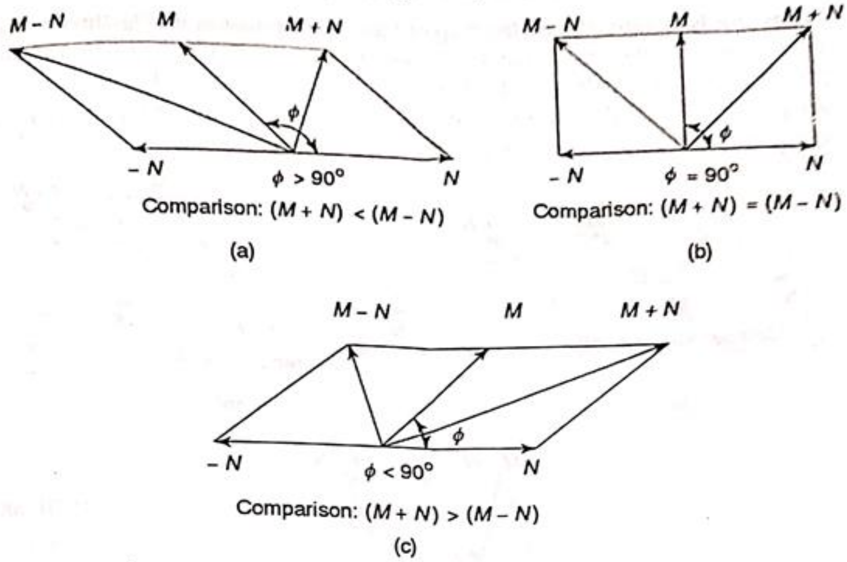


FIGURE 2.17 Vector diagram for amplitude comparator used for phase comparision

Similarly, consider a phase comparator shown in Fig. 2.18 (a). It compares the phases of input signals M and N. If the phase angle between M and N, i.e. angle ϕ is less than 90°, the comparator operates. Now change the input signals to $(M + N)$ and $(M - N)$, as in Fig. 2.18 (b). With these changed inputs, the comparator will operate when phase angle between $(M + N)$ and $(M - N)$, i.e. angle λ is less than 90°. This condition will be satisfied only when $|M| > |N|$. In other words, the phase comparator with changed inputs has now become an

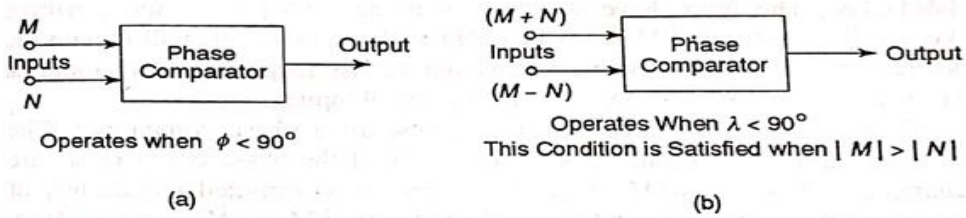


FIGURE 2.18 (a) Phase comparator (b) Phase comparator used for amplitude comparison

amplitude comparator for the original input signals M and N. This has been illustrated with vector diagrams as shown in Fig. 2.19.

Figure (2.17) shows three vector diagrams for an amplitude comparator. The phase angle between the original inputs M and N is ϕ. Now the inputs to the amplitude comparator are changed to $(M + N)$ and $(M - N)$ and its behaviour is examined with the help of three vector diagrams. The three vector diagrams are with phase angel ϕ (*i*) greater than 90°, (ii) equal to 90° and (iii) less than 90°, respectively. When ϕ is less than 90°, $|M + N|$ becomes greater than $|M - N|$ and the relay operates with the modified inputs. When ϕ is equal to 90° or greater than 90°, the relay does not operate.

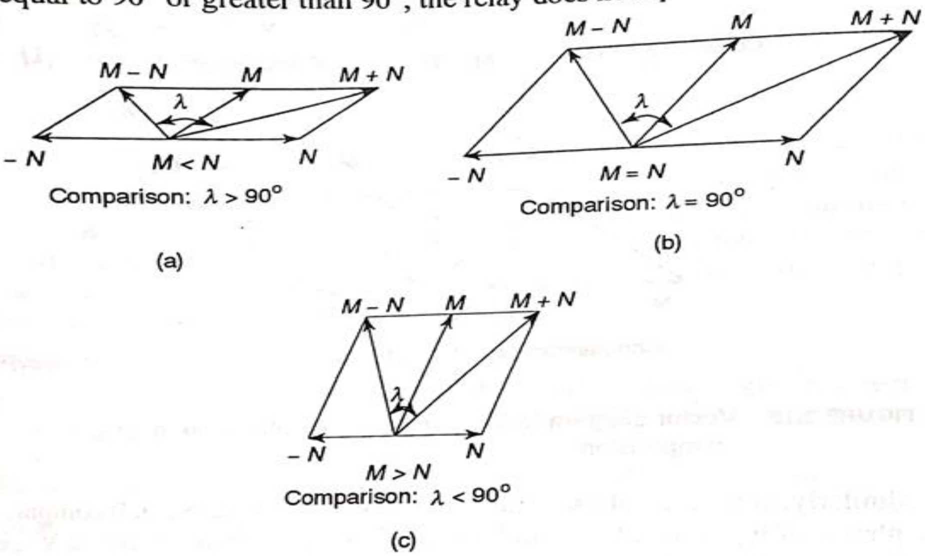


FIGURE 2.19 Phase comparator used for amplitude comparison

The vector diagrams show that $|M+N|$ becomes greater than $|M-N|$ only when ϕ is less than 90°. This will be true irrespective of the magnitude of M and N. In other words, this will be true whether $|M| = |N|$ or $|M| > |N|$ or $|M| < |N|$. The figures have been drawn with $|M| = |N|$. The reader can draw vector diagrams with $|M| < |N|$ or $|M| > |N|$. The results will remain the same. This shows that with changed inputs, the amplitude comparator is

converted to a phase comparator for the original inputs

3.12.2 Instantaneous Overcurrent Relay

The block schematic diagram of the static instantaneous overcurrent relay is shown in Fig. 3.20. The current derived from the main C.T. is fed to the input transformer which gives a proportional output voltage. The input transformer has an air gap in the iron core to give linearity in the current /voltage relationship up to the highest value of current expected, and is provided with tappings on its secondary winding to obtain different current settings. The output voltage of the transformer is rectified through a rectifier and then filtered at a single stage to avoid undesirable time delay in filtering so as to

ensure high speed of operation. A limiter made of a zener diode is also incorporated in the circuit to limit the rectified voltage to safe values even when the input current is very high under fault conditions. A fixed portion of the rectified and filtered voltage (through a potential divider) is compared against a preset pick-up value by a level detector and if it exceeds the pick-up value, a signal through an amplifier is given to the output device which issues the trip signal. The output device may either be a static thyristor circuit or an electro-magnetic slave relay.

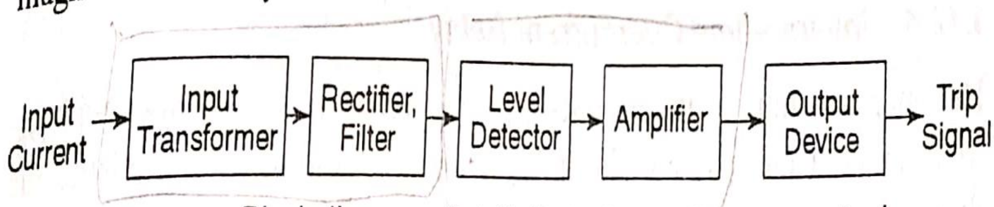


FIGURE 3.20 Block diagram of static instantaneous overcurrent relay

21.14 Distance or Impedance Relays

The operation of the relays discusssed so far depended upon the magnitude of current or power in the protected circuit. However, there is another group of relays in which the operation is governed by the ratio of applied voltage to current in the protected circuit. Such relays are called distance or **impedance relays. In an impedance relay, the torque produced by a current element is opposed by the torque produced by a voltage element. The relay will operate when the ratio V/I is less than a pre-determined value.

Fig. 21.20 illustrates the basic principle of operation of an impedance relay. The voltage element of the relay is excited through a potential transformer (P.T.) from the line to be protected. The current element of the relay is excited from a current transformer (C.T.) in series with the line. The portion AB of the line is the protected zone. Under normal operating conditions, the impedance of the protected zone is Z_L. The relay is so designed that it closes its contacts whenever impedance of the protected section falls below the pre-determined value *i.e.* Z_L in this case.

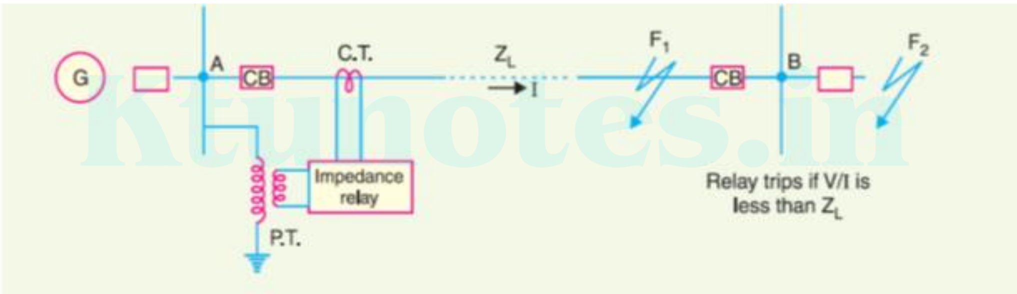


Fig. 21.20

Now suppose a fault occurs at point F_1 in the protected zone. The impedance Z (= $*V/I$) between the point where the relay is installed and the point of fault will be less than Z_L and hence the relay operates. Should the fault occur beyond the protected zone (say point F_2), the impedance Z will be greater than Z_L and the relay does not operate.

Types. A distance or impedance relay is essentially an ohmmeter and operates whenever the impedance of the protected zone falls below a pre-determined value. There are two types of distance relays in use for the protection of power supply, namely ;

(i) *Definite-distance relay* which operates instantaneously for fault upto a pre-determined distance from the relay.

(ii) *Time-distance relay* in which the time of operation is proportional to the distance of fault from the relay point. A fault nearer to the relay will operate it earlier than a fault farther away from the relay.

It may be added here that the distance relays are produced by modifying either of two types of basic relays; the balance beam or the induction disc.

21.15 Definite-Distance Type Impedance Relay

Fig. 21.21 shows the schematic arrangement of a definite-distance type impedance relay. It consists of a pivoted beam F and two electromagnets energised respectively by a current and voltage transformer in the protected circuit. The armatures of the two electromagnets are mechanically coupled to the beam on the opposite sides of the fulcrum. The beam is provided with a bridging piece for the trip contacts. The relay is so designed that the torques produced by the two electromagnets are in the opposite direction.

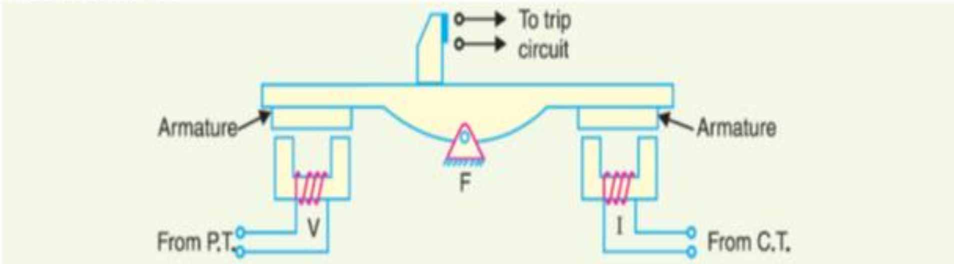


Fig. 21.21

Operation. Under normal operating conditions, the pull due to the voltage element is **greater than that of the current element. Therefore, the relay contacts remain open. However, when a fault occurs in the protected zone, the applied voltage to the relay decreases whereas the current increases. The ratio of voltage to current (*i.e.* impedance) falls below the pre-determined value. Therefore, the pull of the current element will exceed that due to the voltage element and this causes the beam to tilt in a direction to close the trip contacts.

The pull of the current element is proportional to I^2 and that of voltage element to V^2. Consequently, the relay will operate when

$$k_1 V^2 < k_2 I^2$$

or $$\frac{V^2}{I^2} < \frac{k_2}{k_1}$$

or $$\frac{V}{I} < \sqrt{\frac{k_2}{k_1}}$$

or $$Z < \sqrt{\frac{k_2}{k_1}}$$

The value of the constants k_1 and k_2 depends upon the ampere-turns of the two electromagnets. By providing tappings on the coils, the setting value of the relay can be changed.

21.16 Time-Distance Impedance Relay

A time-distance impedance relay is one which automatically adjusts its operating time according to the distance of the relay from the fault point *i.e.*

$$\text{Operating time, } T \propto V/I$$
$$\propto Z$$
$$\propto \text{distance}$$

Construction. Fig. 21.22 shows the schematic arrangement of a typical induction type time-distance impedance relay. It consists of a current driven induction element similar to the double-winding type induction overcurrent relay (refer back to Fig. 21.8). The spindle carrying the disc of this element is connected by means of a spiral spring coupling to a second spindle which carries the bridging piece of the relay trip contacts. The bridge is normally held in the open position by an armature held against the pole face of an electromagnet excited by the voltage of the circuit to be protected.

Operation. Under normal load conditions, the pull of the armature is more than that of the induction element and hence the trip circuit contacts remain open. However, on the occurence of a short-circuit, the disc of the induction current element starts to rotate at a speed depending upon the operating current. As the rotation of the disc proceeds, the spiral spring coupling is wound up till the tension of the spring is sufficient to pull the armature away from the pole face of the *voltage-excited magnet. Immediately this occurs, the spindle carrying the armature and bridging piece moves rapidly in response to the tension of the spring and trip contacts are closed. This opens the circuit breaker to isolate the faulty section.

The speed of rotation of the disc is approximately proportional to the operating current, neglecting the effect of control spring. Also the time of operation of the relay is directly proportional to the pull of the voltage-excited magnet and hence to the line voltage V at the point where the relay is connected. Therefore, the time of operation of relay would vary as V/I *i.e.* as Z or distance.

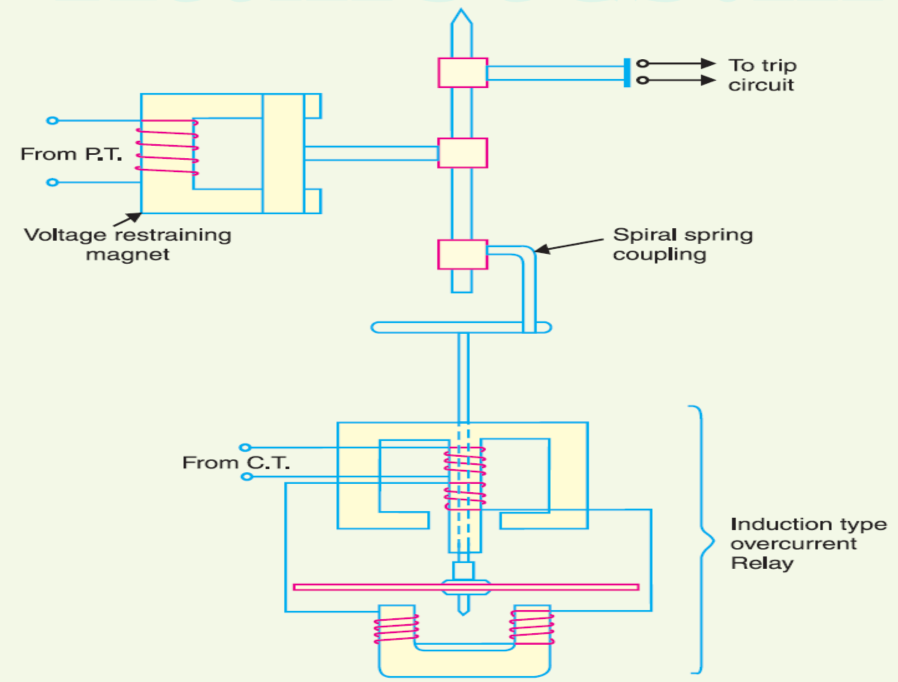


Fig. 21.22

21.17 Differential Relays

Most of the relays discussed so far relied on excess of current for their operation. Such relays are less sensitive because they cannot make correct distinction between heavy load conditions and minor fault conditions. In order to overcome this difficulty, differential relays are used.

A **differential relay** *is one that operates when the phasor difference of two or more similar electrical quantities exceeds a pre-determined value.*

Thus a current differential relay is one that compares the current entering a section of the system with the current leaving the section. Under normal operating conditions, the two currents are equal but as soon as a fault occurs, this condition no longer applies. The difference between the incoming and outgoing currents is arranged to flow through the operating coil of the relay. If this differential current is equal to or greater than the pickup value, the relay will operate and open the circuit breaker to isolate the faulty section.

It may be noted that almost any type of relay when connected in a particular way can be made to operate as a differential relay. In other words, it is not so much the relay construction as the way the relay is connected in a circuit that makes it a differential relay. There are two fundamental systems of differential or balanced protection *viz.*

(*i*) Current balance protection (*ii*) Voltage balance protection

21.18 Current Differential Relay

Fig. 21.23 shows an arrangement of an overcurrent relay connected to operate as a differential relay. A pair of identical current transformers are fitted on either end of the section to be protected (alternator winding in this case). The secondaries of CT's are connected in series in such a way that they carry the induced currents in the same direction. The operating coil of the overcurrent relay is connected across the CT secondary circuit. This differential relay compares the current at the two ends of the alternator winding.

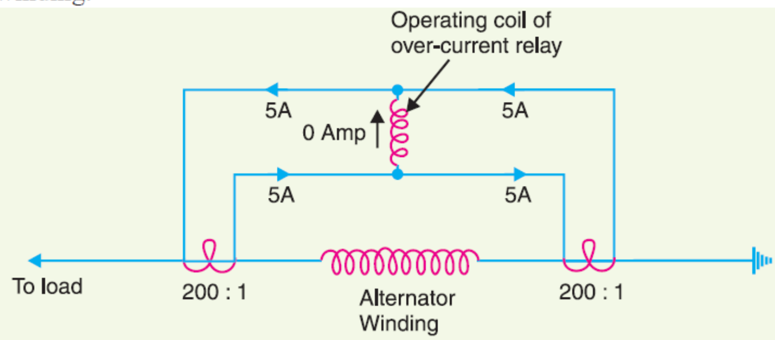


Fig. 21.23

Under normal operating conditions, suppose the alternator winding carries a normal current of 1000 A. Then the currents in the two secondaries of CT's are equal [See Fig. 21.23]. These currents will merely circulate between the two CT's and no current will flow through the differential relay. Therefore, the relay remains inoperative. If a ground fault occurs on the alternator winding as shown in Fig. 21.24 (*i*), the two secondary currents will not be equal and the current flows through the operating coil of the relay, causing the relay to operate. The amount of current flow through the relay will depend upon the way the fault is being fed.

Disadvantages

(*i*) The impedance of the *pilot cables generally causes a slight difference between the currents at the two ends of the section to be protected. If the relay is very sensitive, then the small differential current flowing through the relay may cause it to operate even under no fault conditions.

(*ii*) Pilot cable capacitance causes incorrect operation of the relay when a large through-current flows.

(*iii*) Accurate matching of current transformers cannot be achieved due to pilot circuit impedance.

21.19 Voltage Balance Differential Relay

Fig. 21.27 shows the arrangement of voltage balance protection. In this scheme of protection, two similar current transformers are connected at either end of the element to be protected (*e.g.* an alternator winding) by means of pilot wires. The secondaries of current transformers are connected in series with a relay in such a way that under normal conditions, their induced e.m.f.s' are in opposition.

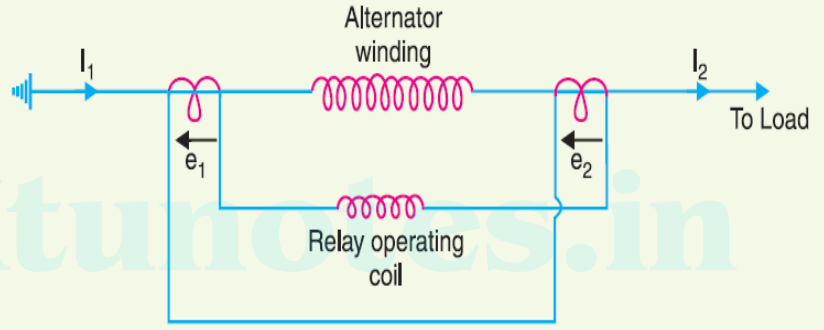


Fig. 21.27

Under healthy conditions, equal currents ($I_1 = I_2$) flow in both primary windings. Therefore, the secondary voltages of the two transformers are balanced against each other and no current will flow through the relay operating coil. When a fault occurs in the protected zone, the currents in the two primaries will differ from one another (*i.e.* $I_1 \neq I_2$) and their secondary voltages will no longer be in balance. This voltage †difference will cause a current to flow through the operating coil of the relay which closes the trip circuit.

Disadvantages

The voltage balance system suffers from the following drawbacks :

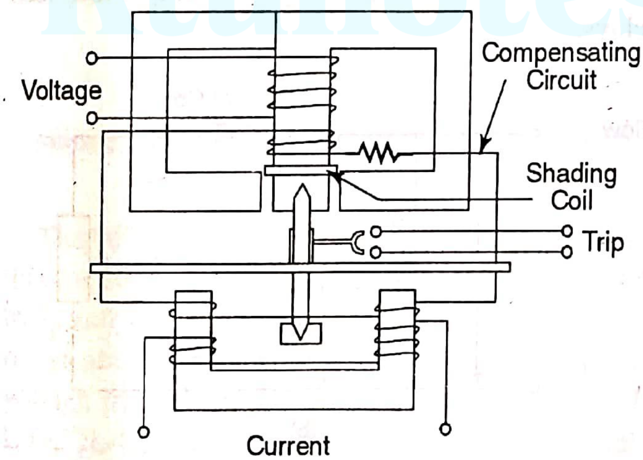
(*i*) A multi-gap transformer construction is required to achieve the accurate balance between current transformer pairs.

(*ii*) The system is suitable for protection of cables of relatively short lengths due to the capacitance of pilot wires. On long cables, the charging current may be sufficient to operate the relay even if a perfect balance of current transformers is attained.

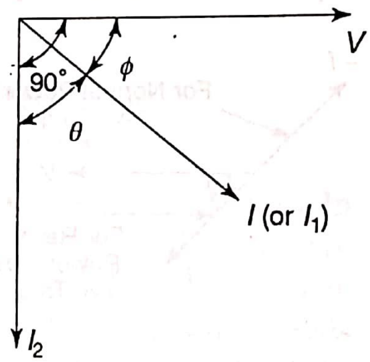
3.5 Reverse Power or Directional Relay

Figure 3.8(a) shows an electromagnetic directional relay. A directional relay is energised by two quantities, namely voltage and current. Fluxes ϕ_1 and ϕ_2 are set up by voltage and current, respectively. Eddy currents induced in the disc by ϕ_1 interact with ϕ_2 and produce a torque. Similarly, ϕ_2 also induces eddy currents in the disc, which interact with ϕ_1 and produce a torque. The resultant torque rotates the disc. The torque is proportional to $VI \cos \phi$, where ϕ is the phase angle between V and I. The torque is maximum when voltage and current are in phase. To produce maximum torque during the fault condition, when the power factor is very poor, a compensating winding and shading are provided, as shown in Fig. 3.8 (a).

Earlier it has been mentioned that the torque produced by an induction relay is given by $T = \phi_1 \phi_2 \sin \theta \propto I_1 I_2 \sin \theta$, where ϕ_1 and ϕ_2 are fluxes produced by I_1 and I_2, respectively. The angle between ϕ_1 and ϕ_2 or I_1 and I_2 is θ. If one of the actuating quantities is voltage, the current flowing in the voltage coil lags behind voltage by approximately 90°. Assume this current to be I_2. The load current I (say I_1) lags V by ϕ. Then the angle θ between I_1 and I_2 is equal to $(90 - \phi)$, as shown in Fig. 3.8(b).



(a) Construction



(b) Phasor diagram

FIGURE 3.8 Induction disc type directional relay

$$T = I_1 I_2 \sin(90 - \phi) \propto I_1 I_2 \cos\phi \propto VI\cos\phi$$

An induction cup construction can also be used to produce a torque proportional to $VI\cos\phi$. The arrangement is shown in Fig. 3.9. Two opposite poles are energised by voltage and the other two poles by current. Here voltage is a polarising quantity. The polarising quantity is one which produces one of the two fluxes. The polarising quantity is taken as a reference with respect to the other quantity which is current in this case.

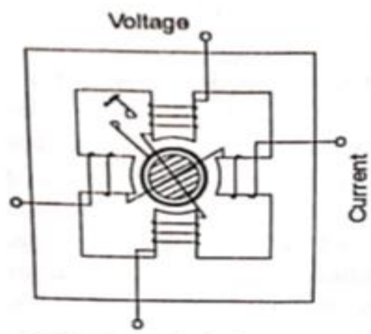


FIGURE 3.9 Induction cup type directional relay

Torque produced is positive when $\cos\phi$ is positive, i.e. ϕ is less than 90°. When ϕ is more than 90° (between 90° and 180°), the torque is negative. At a particular relay location, when power flows in the normal direction, the relay is connected to produce negative torque. The angle between the actuating quantities supplied to the relays is kept $(180° - \phi)$ to produce negative torque. If due to any reason, the power flows in the reverse direction, the relay produces a positive torque and it operates. In this condition, the angle between the actuating quantities ϕ is kept less than 90° to produce a positive torque. This is shown in Fig. 3.10 (a). For normal flow of power, the relay is supplied with V and $-I$. For reverse flow, the actuating quantities become V and I. Torque becomes $VI\cos\phi$, i.e. positive. This can be achieved easily by reversing the current coil, as shown in Fig. 3.10 (b).

Relaying units supplied with single actuating quantity discussed earlier are non-directional overcurrent relays. Non-directional relays are simple and less expensive than directional relays.

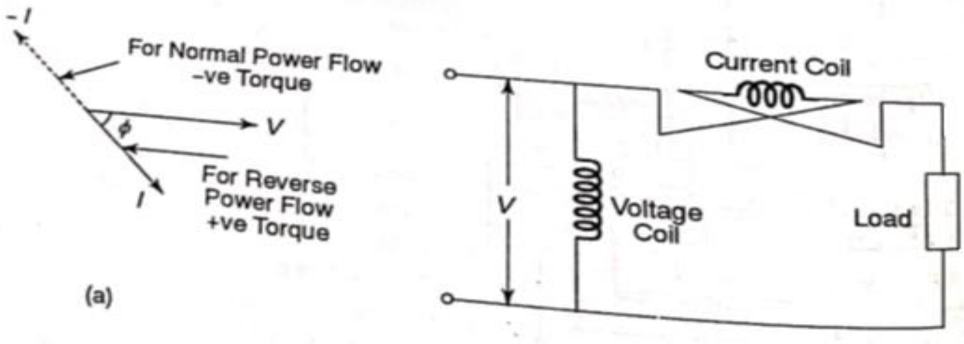


FIGURE 3.10 (a) Phasor diagram for directional relay
(b) Connection of current coil for reverse power relay

MICROPROCESSOR BASED OVER CURRENT RELAY

Modern power networks require faster, more accurate and reliable protective schemes. Microprocessor based protective schemes are capable of fulfilling these requirements. They are superior to electromagnetic and static relays. These schemes have more flexibility due to their programmable approach when compared with static relays which have hard wired circuitry. Microprocessor based schemes are more compact, accurate and fast.

<u>Microprocessor Based Overcurrent Relay</u>

An over current relay is the simplest form of protective relay which operates when the current in any circuit exceeds a certain predetermined value ie the pick up value. Using a multiplexer the microprocessor can sense the fault current of a number of circuits. If the fault current in any circuit exceeds pickup value the microprocessor sends a tripping sinal to the circuit breaker of the faulty circuit.

Micro processor accepts signal in voltage form. So current is taken from C.T. and given to I to V converter because many electronics circuit require voltage signal for operation.

The A.C. voltage is converted into D.C. voltage by using rectifier. This D.C. voltage is proportional to load current only. The output of rectifier is given to Multiplexer.

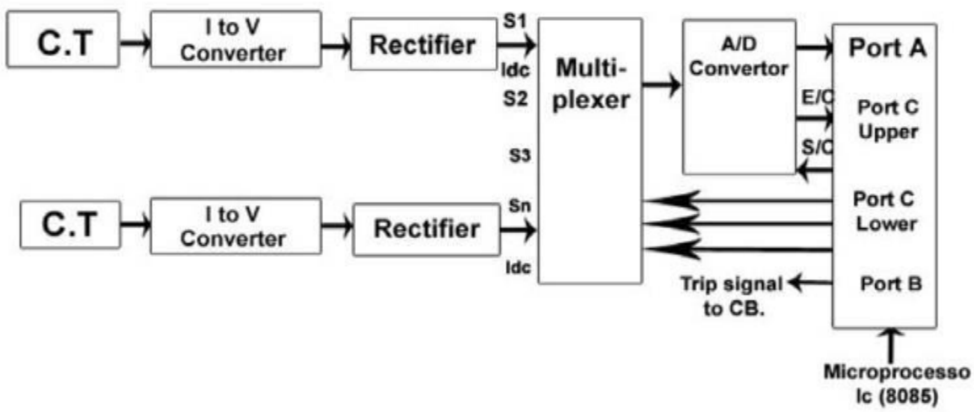
The Multiplexer gives output to A/D Converter where Analog DC voltage is converted to Digital form (in form of 0 and 1 i.e. binary form). Microprocessor understands only codes in 0 and 1 form.

μP (Microprocessor) gives S/C (start of conversion) signal to A/D converter (i.e. analog to digital conversion is started and μP gives permission to A/D convertor for this by sending S/C)

When converting from analog to digital is over (finish) then A/D converter sends E/C signal to μP (E/C – End of Conversion).

When work of A/D is over then μp compare the magnitude of this incoming current with required current value (i.e. set value or reference value).

If incoming value is more – fault is occurring and trip signal is sent to CB circuit breaker.



Microprocessor Based Over Current Relay

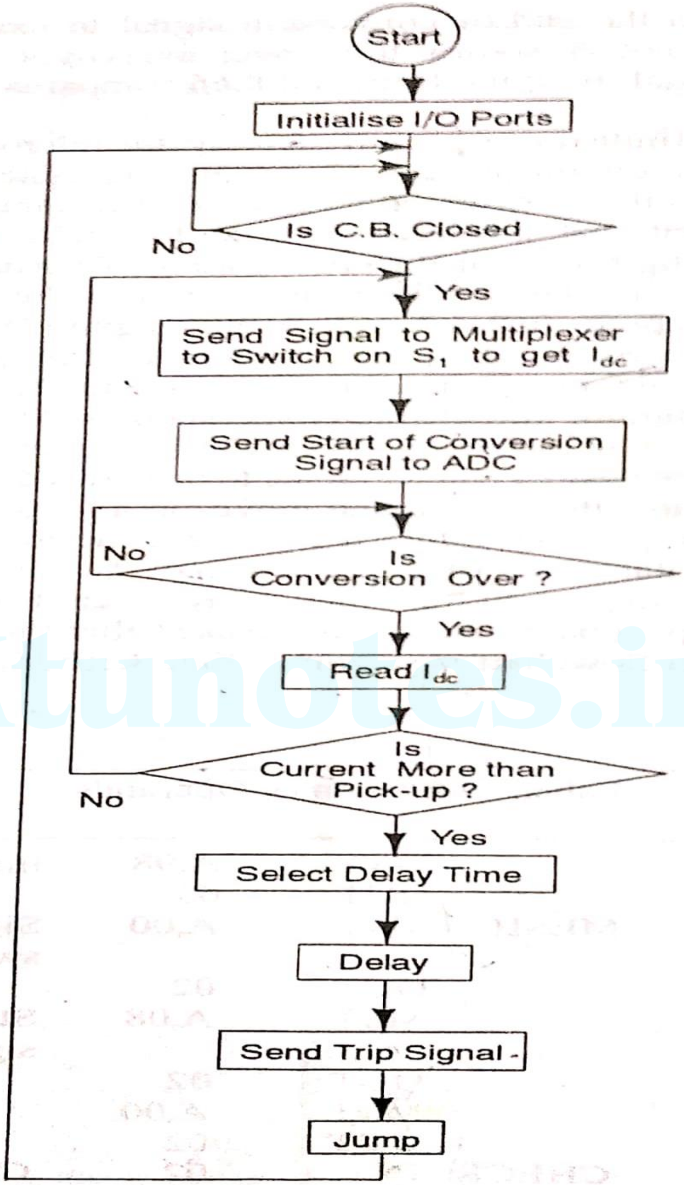


FIGURE 8.1(b) Program flowchart for overcurrent relay

Numerical Relay

Why Numerical Relaying?

Numerical relays are based on digital technology

To improve reliability.

In turn, increase in dependability as well as security

By offering very low impedance to C.T secondary, reducing burden (voltamperes) of Current Transformer (CT) and Voltage Transformer (VT)

A computer relay can be programmed.

Due to the programming feature, it is possible to have generic hardware for multiple relays, which reduces the cost of inventory.

Magnitude scaling and phase shift to a voltage signal to generate line to line voltage from phase to neutral voltage is much simpler with computer aided relaying because it can be handled by the program. Developments in fibre optic communication have pioneered development in numerical relay. Once, the analogue signals from CTs and VTs are digitized, they can be converted to optical signals and transmitted on substation LAN using fibre optic network

Numerical relays can be nicely interfaced with a substation LAN

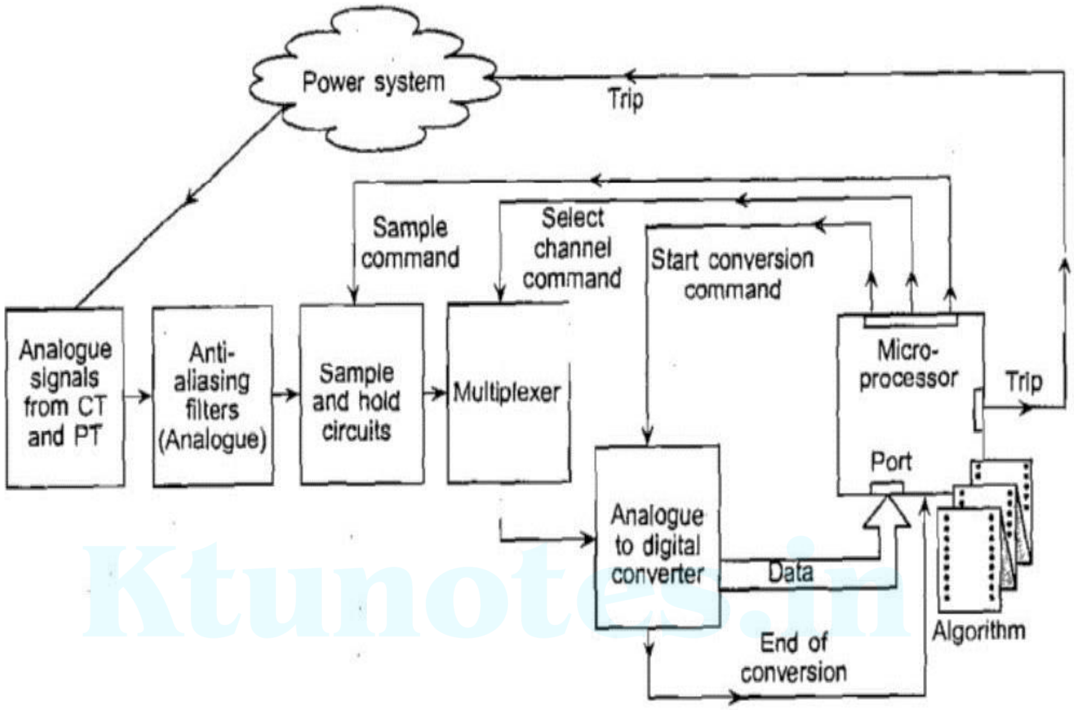
Numerical relay is a device in which measured electrical quantities are sequentially sampled and then converted into numerical data which are mathematically or logically processed to take decision for issuing trip command.

It is based on numerical (digital) devices like microprocessors, microcontrollers, digital signal processors (DSPs) etc. this relay acquires sequential samples of the ac quantities in numeric (digital) data form through the data acquisition system (DAS). Ina numeric relay, the analog current and voltage signals monitored through primary transducers (CTs and VTs) are conditioned, sampled at specified instants of time and converted to digital form for numerical manipulation,, analysis, display and recording.

The signal conditioner electrically isolates the relay from the power system, reduces the level of the input voltages, converts currents to equivalent voltages and removes high frequency components from the signals using analog filters. The relay is isolated from the power system by using auxiliary transformers which receive analog signals and reduce their levels to make them suitable for use in relays. Since the A/D converters accept voltage signals only, the current signals are converted into proportional voltage signals by using I/V converters or by passing through precision shunt resistors. Anti-aliasing filters (which are low-pass filters) are used to prevent aliasing from affecting relaying functions.

The outputs of the signal conditioner (the analog input subsystem) are applied to the analog interface, which includes sample hold (S/H) circuits, analog multiplexers and analog-to-digital (A/D) converters. These components sample the reduced level signals and convert their analog levels to equivalent numbers that are stored in memory. The status of isolators and circuit breakers in the power system is provided to the relay via the digital input subsystem and are read into the microcomputer/microcontroller memory.

After quantization by the A/D converter, analog electrical signals are represented by discrete values of the samples taken at specified instants of time. The signals in the form of discrete numbers are processed by a relaying algorithm using numerical methods. A relaying algorithm which processes the acquired information is a part of the software. The algorithm uses signal-processing technique to estimate the real and imaginary components of fundamental frequency voltage and current phasor.



Important Terms

It is desirable to define and explain some important terms much used in connection with relays.

(*i*) Pick-up current. It is the minimum current in the relay coil at which the relay starts to operate. So long as the current in the relay is less than the pick-up value, the relay does not operate and the breaker controlled by it remains in the closed position. However, when the relay coil current is equal to or greater than the pickup value, the relay operates to energise the trip coil which opens the circuit breaker.

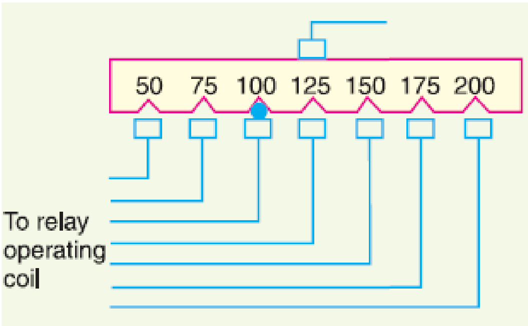


Fig. 21.14

(*ii*) Current setting. It is often desirable to adjust the pick-up current to any required value. This is known as current setting and is usually achieved by the use of tappings on the relay operating coil. The taps are brought out to a plug bridge as shown in Fig. 21.14. The plug bridge permits to alter the number of turns on the relay coil. This changes the torque on the disc and hence the time of operation of the relay. The values assigned to each tap are expressed in terms of percentage full-load rating of C.T. with which the relay is associated and represents the value *above* which the disc commences to rotate and finally closes the trip circuit.

Pick-up current = Rated secondary current of C.T. ∗ Current setting

For example, suppose that an overcurrent relay having current setting of 125% is connected to a supply circuit through a current transformer of 400/5. The rated secondary current of C.T. is 5 amperes. Therefore, the pick-up value will be 25% more than 5 A *i.e*. 5 · 1·25 = 6·25 A. It means that with above current setting, the relay will actually operate for a relay coil current equal to or greater than 6·25 A.

The current plug settings usually range from 50% to 200% in steps of 25% for overcurrent relays and 10% to 70% in steps of 10% for earth leakage relays. The desired current setting is obtained by inserting a plug between the jaws of a bridge type socket at the tap value required.

(*iii*) Plug-setting multiplier (P.S.M.). It is the ratio of fault current in relay coil to the pick-up current *i.e*. P.S.M. =Fault current in relay coil / Pick - up current

$$= \frac{\text{Fault current in relay coil}}{\text{Rated secondary current of } CT \times \text{Current setting}}$$

For example, suppose that a relay is connected to a 400/5 current transformer and set at 150%. With a primary fault current of 2400 A, the plug-setting multiplier can be calculated as under :

$$\text{Pick-up value} = \text{Rated secondary current of } CT \times \text{Current setting}$$
$$= 5 \times 1{\cdot}5 = 7{\cdot}5 \text{ A}$$
$$\text{Fault current in relay coil} = 2400 \times \frac{5}{400} = 30 \text{ A}$$
$$\therefore \quad \text{P.S.M.} = 30/7{\cdot}5 = 4$$

(*iv*) Time-setting multiplier. A relay is generally provided with control to adjust the time of operation. This adjustment is known as time-setting multiplier. The time-setting dial is calibrated from 0 to 1 in steps of 0.05 sec (see Fig. 21.15). These figures are multipliers to be

used to convert the time derived from time/P.S.M. curve into the actual operating time. Thus if the time setting is 0·1 and the time obtained from the time/P.S.M. curve is 3 seconds, then actual relay operating time = 3 ∗ 0·1 = 0·3 second. For instance, in an induction relay, the time of operation is controlled by adjusting the amount of travel of the disc from its reset position to its pickup position. This is achieved by the adjustment of the position of a movable backstop which controls the travel of the disc and thereby varies the time in which the relay will close its contacts for given values of fault current. A so-called "time dial" with an evenly divided scale provides this adjustment. The acutal time of operation is calculated by multiplying the time setting multiplier with the time obtained from time/P.S.M. curve of the relay.

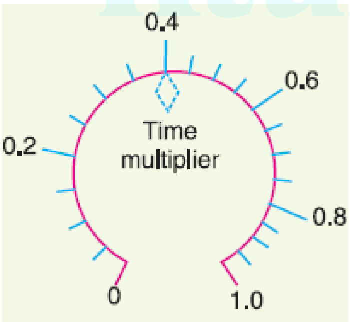


Fig. 21.15

Time/P.S.M. Curve

Fig. 21.16 shows the curve between time of operation and plug setting multiplier of a typical relay. The horizontal scale is marked in terms of plug-setting multiplier and represents the number of times the relay current is in excess of the current setting. The vertical scale is marked in terms of the time required for relay operation. If the P.S.M. is 10, then the time of operation (from the curve) is 3 seconds. The actual time of operation is obtained by multiplying this time by the time-setting multiplier. It is evident from Fig. 21.16 that for lower values of overcurrent, time of operation varies inversely with the current but as the current approaches 20 times full-load value, the operating time of relay tends to become constant. This feature is necessary in order to ensure discrimination on very heavy fault currents flowing through sound feeders.

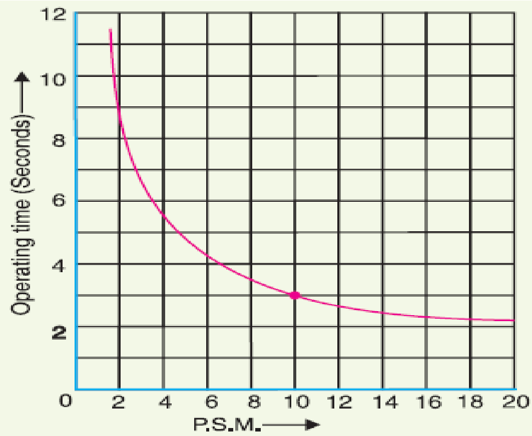


Fig. 21.16

21.9 Calculation of Relay Operating Time

In order to calculate the actual relay operating time, the following things must be known :

(a) Time/P.S.M. curve
(b) Current setting
(c) Time setting
(d) Fault current
(e) Current transformer ratio

The procedure for calculating the actual relay operating time is as follows :

(i) Convert the fault current into the relay coil current by using the current transformer ratio.
(ii) Express the relay current as a multiple of current setting *i.e.* calculate the P.S.M.
(iii) From the Time/P.S.M. curve of the relay, read off the time of operation for the calculated P.S.M.
(iv) Determine the actual time of operation by multiplying the above time of the relay by time-setting multiplier in use.

Example 21.1. *Determine the time of operation of a 5-ampere, 3-second overcurrent relay having a current setting of 125% and a time setting multiplier of 0·6 connected to supply circuit through a 400/5 current transformer when the circuit carries a fault current of 4000 A. Use the curve shown in Fig. 21.16.*

Solution.

$$\text{Rated secondary current of C.T.} = 5\text{ A}$$

$$\text{Pickup current} = 5 \times 1{\cdot}25 = 6{\cdot}25\text{ A}$$

$$\text{Fault current in relay coil} = 4000 \times \frac{5}{400} = 50\text{ A}$$

$$\therefore \quad \text{Plug-setting multiplier (P.S.M.)} = \frac{50}{6 \cdot 25} = 8$$

Corresponding to the plug-setting multiplier of 8 (See Fig. 21.16), the time of operation is 3.5 seconds.

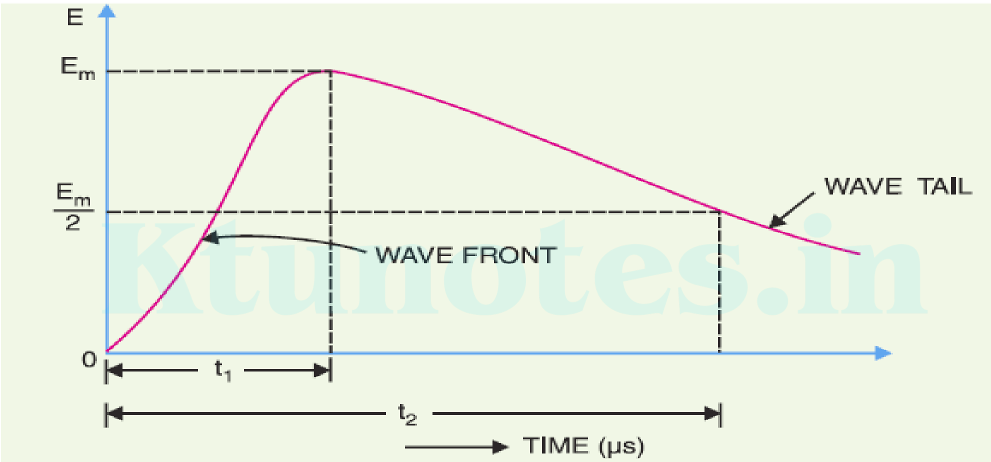
$$\therefore \quad \text{Actual relay operating time} = 3{\cdot}5 \times \text{Time-setting} = 3{\cdot}5 \times 0{\cdot}6 = \mathbf{2{\cdot}1\text{ seconds}}$$

Voltage surge

A sudden rise in voltage for a very short duration on the power system is known as a **voltage surge** *or* **transient voltage.**

Transients or surges are of temporary nature and exist for a very short duration (a few hundred μ*s*) but they cause over voltages on the power system. They originate from switching and from other causes but by far the most important transients are those caused by lightning striking a transmission line. When lightning strikes a line, the surge rushes along the line, just as a flood of water rushes along a narrow valley when the retaining wall of a reservoir at its head suddenly gives way. In most of the cases, such surges may cause the line insulators (near the point where lightning has struck) to flash over and may also damage the nearby transformers, generators or other equipment connected to the line if the equipment is not suitably protected.

Figure shows the wave-form of a typical lightning surge. The voltage build-up is taken along *y*-axis and the time along *x*-axis. It may be seen that lightning introduces a steep-fronted wave. The steeper the wave front, the more rapid is the build-up of voltage at any point in the network. In most of the cases, this build-up is comparatively rapid, being of the order of 1–5 μs. Voltage surges are generally specified in terms of *rise time *t*1 and the time *t*2 to decay to half of the peak value.



CAUSES OF OVER VOLTAGES

The overvoltages on a power system may be broadly divided into two main categories *viz*.

1. Internal causes

(*i*) Switching surges **(*ii*)** Insulation failure **(*iii*)** Arcing ground **(*iv*)** Resonance

2. External causes *i.e*. lightning

Internal causes do not produce surges of large magnitude. Generally, surges due to internal causes are taken care of by providing proper insulation to the equipment in the power system. However, surges due to lightning are very severe and may increase the system voltage to several times the normal value. If the equipment in the power system is not protected against lightning surges, these surges may cause considerable damage. In fact, in a power system, the protective devices provided against over voltages mainly take care of lightning surges.

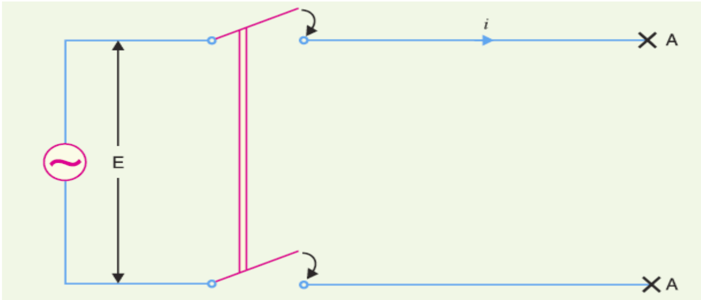
INTERNAL CAUSES OF OVERVOLTAGE

Internal causes of over voltages on the power system are primarily due to oscillations set up by the sudden changes in the circuit conditions. This circuit change may be a normal switching operation such as opening of a circuit breaker, or it may be the fault condition such as grounding of a line conductor.

Internal causes of over voltages.

1. Switching Surges. The over voltages produced on the power system due to switching operations are known as switching surges. A few cases are :

(*i*) Case of an open line. During switching operations of an unloaded line, travelling waves are set up which produce over voltages on the line. Consider an unloaded line being connected to a voltage source as shown in Figure.

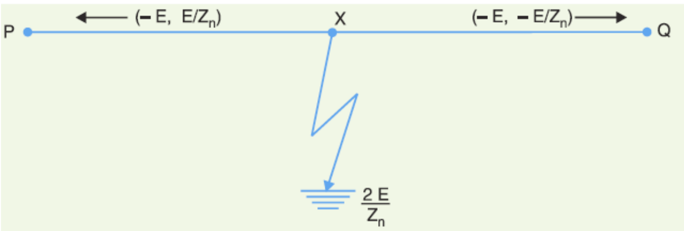


When the unloaded line is connected to the voltage source, a voltage wave is set up which travels along the line. On reaching the terminal point *A*, it is reflected back to the supply end without change of sign. This causes voltage doubling *i.e*. voltage on the line becomes twice the normal value. If $E_{r.m.s.}$ is the supply voltage, then instantaneous voltage which the line will have to withstand will be $2\sqrt{2}\ E$. This overvoltage is of temporary nature.

(*ii*) Case of a loaded line. Over voltages will also be produced during the switching operations of a loaded line. Suppose a loaded line is suddenly interrupted. This will set up a voltage of 2 Z_n *I* across the break (*i.e*. switch) where *i* is the instantaneous value of current at the time of opening of line and Z_n is the natural impedance of the line.

(*iii*) Current chopping. Current chopping results in the production of high voltage transients across the contacts of the air blast circuit breaker. When breaking low currents (*e.g.* transformer magnetising current) with air-blast breaker, the powerful de-ionising effect of air-blast causes the current to fall abruptly to zero well before the natural current zero is reached. This phenomenon is called current chopping and produces high transient voltage across the breaker contacts. Over voltages due to current chopping are prevented by resistance switching.

2. Insulation failure. The most common case of insulation failure in a power system is the grounding of conductor (*i.e*. insulation failure between line and earth) which may cause over voltages in the system.



Suppose a line at potential *E* is earthed at point *X*. The earthing of the line causes two equal voltages of −*E* to travel along *XQ* and *XP* containing currents $-E/Z_n$ and $+E/Z_n$ respectively. Both these currents pass through *X* to earth so that current to earth is $2\ E/Z_n$.

3. Arcing ground. The arcing ground produces severe oscillations of three to four times the normal voltage. *The phenomenon of intermittent arc taking place in line-to-ground fault of a 3ϕ system with consequent production of transients is known as* **arcing ground.** The transients produced due to arcing ground are cumulative and may cause serious damage to the equipment in the power system by causing breakdown of insulation. Arcing ground can be prevented by earthing the neutral.

4. Resonance. Resonance in an electrical system occurs when inductive reactance of the circuit becomes equal to capacitive reactance. Under resonance, the impedance of the circuit is equal to resistance of the circuit and the p.f. is unity. Resonance causes high voltages in the electrical system. In the usual transmission lines, the capacitance is very small so that resonance rarely occurs at the fundamental supply frequency. However, if generator *e.m.f.* wave is distorted, the trouble of resonance may occur due to 5th or higher harmonics and in case of underground cables too.

External cause of overvoltages: lightning

An electric discharge between cloud and earth, between clouds or between the charge centres of the same cloud is known as **lightning.**

Lightning is a huge spark and takes place when clouds are charged to such a high potential (+ve or −ve) with respect to earth or a neighbouring cloud that the dielectric strength of neighbouring medium (air) is destroyed.

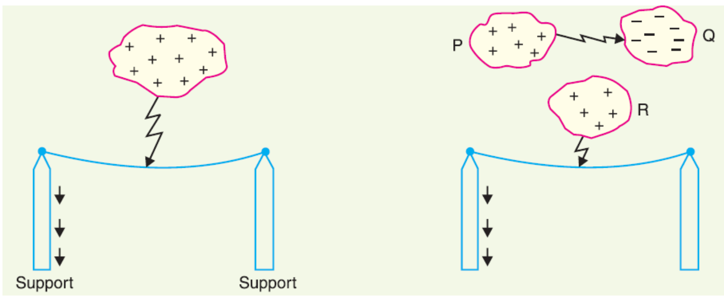
Types of Lightning Strokes **1.** Direct stroke **2.** Indirect stroke

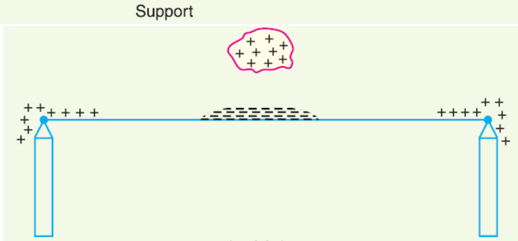
1. Direct stroke. In the direct stroke, the lightning discharge (*i.e.* current path) is directly from the cloud to the subject equipment *e.g*. an overhead line. From the line, the current path may be over the insulators down the pole to the ground. The over voltages set up due to the stroke may be large enough to flashover this path directly to the ground. The direct strokes can be of two types *viz*.

(*i*) Stroke *A* and (*ii*) stroke *B*.

In stroke *A*, the lightning discharge is from the cloud to the subject equipment *i.e*. an overhead line in this case as shown in Fig i In stroke *B*, the lightning discharge occurs on the overhead line as a result of stroke *A* between the clouds as shown in Fig ii

2. Indirect stroke. Indirect strokes result from the electrostatically induced charges on the conductors due to the presence of charged clouds. This is illustrated in Fig. iii A positively charged cloud is above the line and induces a negative charge on the line by electrostatic induction. This negative charge, however, will be only on that portion of the line right under the cloud and the portions of the line away from it will be positively charged as shown in Fig.iii. The induced positive charge leaks slowly to earth *via* the insulators. When the cloud discharges to earth or to another cloud, the negative charge on the wire is isolated as it cannot flow quickly to earth over the insulators. The result is that negative charge rushes along the line is both directions in the form of travelling waves.





SURGES OR OVER VOLTAGES

Surges, or transients, are brief overvoltage spikes or disturbances on a power waveform that can damage, degrade, or destroy electronic equipment within any home, commercial building, or manufacturing facility.

SURGE DIVERTER

A **lightning arrester** *or a* **surge diverter** *is a protective device which conducts the high voltage surges on the power system to the ground,* thus protecting sensitive electrical and electronic equipment.

It consists of a spark gap in series with a non-linear resistor. One end of the diverter is connected to the terminal of the equipment to be protected and the other end is effectively grounded. The length of the gap is so set that normal line voltage is not enough to cause an arc across the gap but a dangerously high voltage will break down the air insulation and form an arc. The property of the non-linear resistance is that its resistance decreases as the voltage (or current) increases and vice-versa.

(*i*) Under normal operation, the lightning arrester is off the line *i.e.* it conducts no current to earth or the gap is non-conducting.

(*ii*) On the occurrence of overvoltage, the air insulation across the gap breaks down and an arc is formed, providing a low resistance path for the surge to the ground. In this way, the excess charge on the line due to the surge is harmlessly conducted through the arrester to the ground instead of being sent back over the line.

TYPES OF SURGE DIVERTER

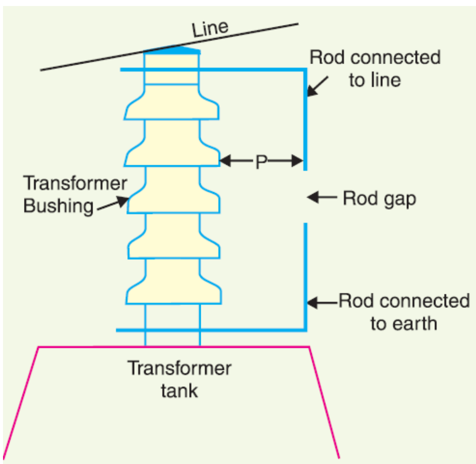
Rod gap, Horn gap, Protector tube or expulsion type surge diverter , Valve type surge diverter

ROD GAP

It is a very simple type of diverter and consists of two 1.5 cm rods, which are bent at right angles with a gap in between. One rod is connected to the line circuit and the other rod is connected to earth.

The distance between gap and insulator must not be less than one third of the gap length so that the arc may not reach the insulator and damage it.

Under normal operating conditions, the gap remains non-conducting. On the occurrence of a high voltage surge on the line, the gap sparks over and the surge current is conducted to earth. In this way excess charge on the line due to the surge is harmlessly conducted to earth.



LIMITATIONS

After the surge is over, the arc in the gap is maintained by the normal supply voltage, leading to short-circuit on the system.

The rods may melt or get damaged due to excessive heat produced by the arc.

The climatic conditions (e.g. rain, humidity, temperature etc.) affect the performance of rod gap arrester.

The polarity of the surge also affects the performance of this arrester.

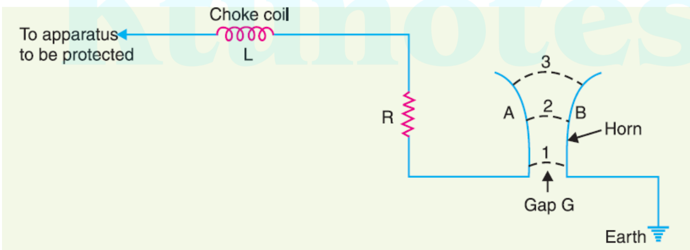
HORN GAP ARRESTOR

It consists of two horn shaped metal rods *A* and *B* separated by a small air gap. The horns are so constructed that distance between them gradually increases towards the top as shown.

One end of horn is connected to the line through a resistance *R* and choke coil *L* while the other end is effectively grounded. The resistance *R* helps in limiting the follow current to a small value.

The choke coil is so designed that it offers small reactance at normal power frequency but a very high reactance at transient frequency. Thus the choke does not allow the transients to enter the apparatus to be protected. The gap between the horns is so adjusted that normal supply voltage is not enough to cause an arc across the gap.

Under normal conditions, the gap is non-conducting *i.e*. normal supply voltage is insufficient to initiate the arc between the gap. On the occurrence of an overvoltage, spark-over takes place across the small gap *G*. The heated air around the arc and the magnetic effect of the arc cause the arc to travel up the gap. The arc moves progressively into positions 1, 2 and 3. At some position of the arc (perhaps position 3), the distance may be too great for the voltage to maintain the arc. Consequently, the arc is extinguished. The excess charge on the line is thus conducted through the arrester to the ground.



EXPULSION TYPE SURGE DIVERTER

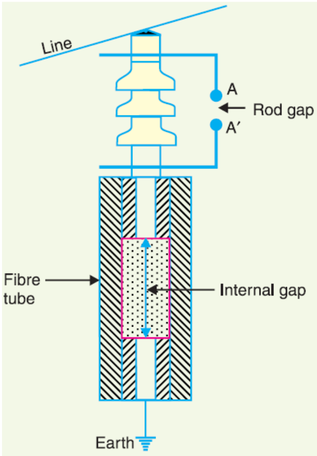
This type of arrester is also called ‘protector tube’ and is commonly used on system operating at voltages up to 33kV.

It essentially consists of a rod gap in series with a second gap enclosed within the fiber tube. The gap in the fiber tube is formed by two electrodes.

The upper electrode is connected to rod gap and the lower electrode to the earth.

On the occurrence of an overvoltage on the line, the series gap *A A′* is spanned and an arc is struck between the electrodes in the tube. The heat of the arc vaporises some of the fibre of tube walls, resulting in the production of a neutral gas.

In an extremely short time, the gas builds up high pressure and is expelled through the lower electrode which is hollow. As the gas leaves the tube violently, it carries away ionised air around the arc. This de-ionising effect is generally so strong that arc goes out at a current zero and will not be re-established.



ADVANTAGES

- They are not very expensive.
- They can be easily installed.
- They are improved form of rod gap arresters as they block the flow of power frequency follow currents.

LIMITATIONS

- An expulsion type arrester can perform only limited number of operations as during each operation some of the fiber material is used up.
- This type of arrester cannot be mounted on enclosed equipment due to discharge of gases during operation.
- Due to the poor volt/amp characteristic of the arrester, it is not suitable for protection of expensive equipment.

VALVE TYPE ARRESTERS

Valve type arresters incorporate non linear resistors and are extensively used on systems, operating at high voltages.

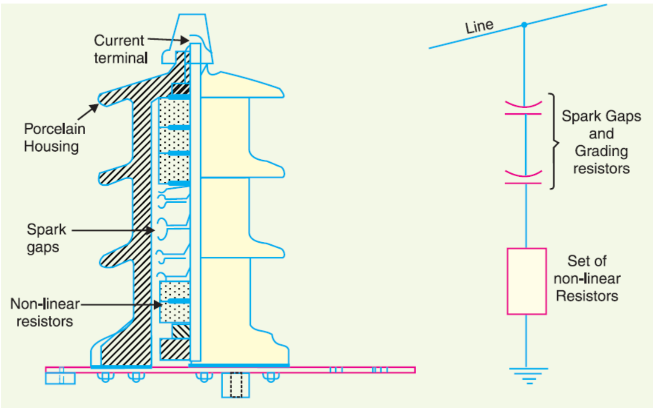
It consists of two assemblies (i) series spark gaps and (ii) non-linear resistor discs

The non-linear elements are connected in series with the spark gaps. Both the assemblies are accommodated in tight porcelain container.

The spark gap is a multiple assembly consisting of a number of identical spark gaps in series. Each gap consists of two electrodes with fixed gap spacing. The spacing of the series gaps is such that it will withstand the normal circuit voltage.

An over voltage will cause the gap to break down causing the surge current to ground via the non-linear resistors. The non-linear resistors have the property of offering a high resistance to current flow when normal system voltage is applied, but a low resistance to the flow of high surge currents. When the surge is over the non linear resistor assume high resistance to stop the flow of current.

Under normal conditions, the normal system voltage is insufficient to cause the breakdown of air gap assembly. On the occurrence of an over voltage, the breakdown of the series spark gap takes place and the surge current is conducted to earth via the nonlinear resistances. Since the magnitude of surge current is very large, the nonlinear elements will offer a very low resistance to the passage of surge. The surge will rapidly go to earth instead of being sent back over the line.



ADVANTAGES

- They provide very effective protection against surges.
- They operate very rapidly taking less than a second
- The impulse ratio is practically unity.

LIMITATIONS

- They may fail to check the surge of very steep wave front reaching the terminal apparatus. This calls for additional steps to check steep fronted waves.
- Their performance is adversely affected by the entry of moisture into the enclosure. This necessitates effective sealing of the enclosure at all times.

11.10 Insulation Coordination

Insulation coordination is the correlation of the insulation of electrical equipment and lines with the characteristics of protective devices such that the insulation of the whole power system is protected from excessive overvoltages.

The main aim of insulation coordination is the selection of suitable values for the insulation level of the different components in any power system and their arrangement in a reasonable manner so that the whole power system is protected from overvoltages of excessive magnitude. Thus, the insulation strength of various equipment, like transformers, circuit breakers, etc. should be higher than that of the lightning arresters and other surge protective devices. The insulation coordination is thus the matching of the volt-time flashover and breakdown characteristics of equipment and protective devices, in order to obtain maximum protective margin at a reasonable cost. The volt-time curves of equipment to be protected and the protective device are shown in Fig. 11.24. Curve A is the volt-time curve of the protective device and curve B is the volt-time curve of the equipment to be protected. From volt-time curves A and B of Fig. 11.24, it is clear that any insulation having a voltage withstanding strength in excess of the insulation strength of curve B will be protected by the protective device of curve A.

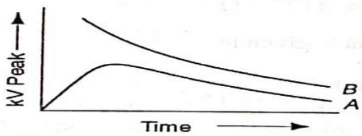
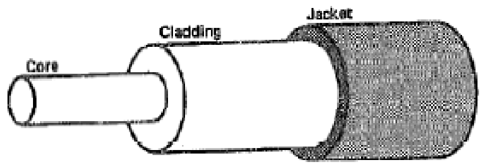


FIGURE 11.24 Volt-time curves of protective device and the equipment to be protected

Fiber Optic Communication

Fiber optic communication is a method of transmitting data using light, rather than electricity like standard wires and cabling. Fiber optic cables are used in different areas with the largest being for telephones, the Internet, and television. It uses total internal reflection principle to transmit light.

Fiber optics (optical fibers) are long, thin strands of very pure glass about the size of a human hair. They are arranged in bundles called optical cables and used to transmit signals over long distances.



Every fiber optic system is composed of three main parts:

- Transmitter
- Fiber optic cable
- Receiver

The transmitter takes analog or digital information and converts it into digital pulses that are then transmitted through a light source. This light source is connected to the fiber optic cable which carries the signal in the form of light pulses to the receiver. The receiver has a light detector that reads the incoming light pulses and amplifies them so that a digital or analog signal can received

Optical power transmission

Optical fibers or fiber cables can be used for transmitting optical power from a source to some application. The term power over fiber or photonic power implies that the optical power is generated from electric power – usually with a laser diode – and at the end converted back to electrical power for some electronic device. That conversion can be done with a photovoltaic cell, i.e., a semiconductor device based on a material such as gallium arsenide, indium phosphide, or indium gallium arsenide.

Fiber Optics Transmission

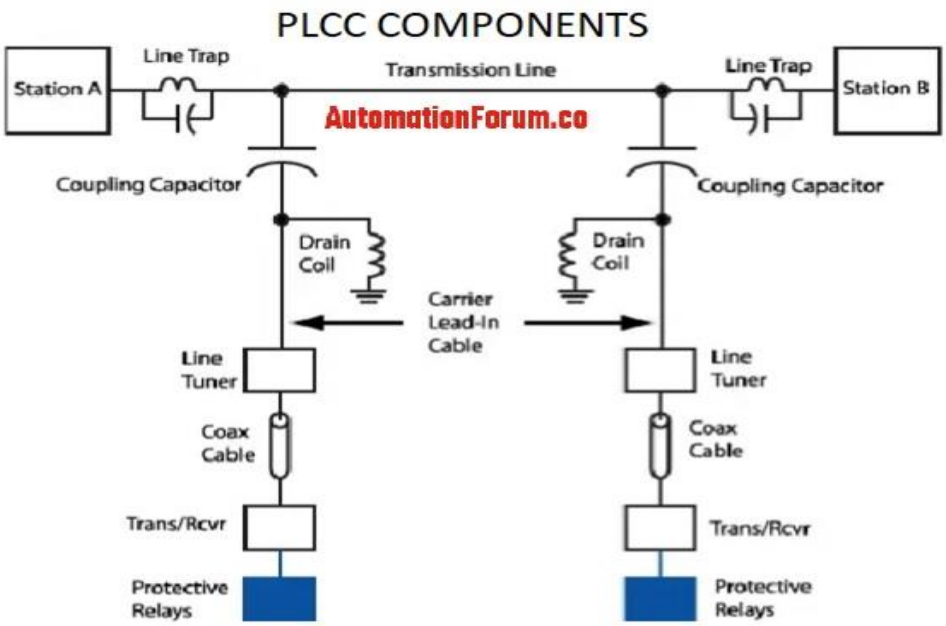
- Low Attenuation
- Very High Bandwidth (THz)
- Small Size and Low Weight
- No Electromagnetic Interference
- Low Security Risk
- Elements of Optical Transmission
  - Electrical-to-optical Transducers
  - Optical Media
  - Optical-to-electrical Transducers
  - Digital Signal Processing, repeaters and clock recovery.

PLCC: POWER LINE CARRIER COMMUNICATION

In the early 20th century the power companies used telephones as the medium of communication for exchange of voice messages for operational support, maintenance, control etc and as a method of connectivity at remote locations. The telephone lines ran parallel to the power lines. This had so many disadvantages:

- The use of telephone circuits over large distances and at difficult terrains like mountains was very expensive.
- Noise interference due to currents flowing in parallel power lines over the telephone circuits.
- Frequent shut down of telephone cables during harsh weather conditions like snows in winter, storms etc made them less reliable.

This led to the idea of inventing a more robust and less expensive method of communication. The use of power line as a method of telephony was a long thought idea and its first successful test took place in Japan in 1918. And there after its commercialization started during 1930s.



PLCC system consists of three parts:

1. The terminal assemblies include the receiver transmitter and protective relays.
2. The coupling equipment is the combination of line tuner, coupling capacitor and the wave or line trap.
3. The 50/60 Hz power transmission line serves as path for relaying data in the PLCC bandwidth.

Coupling Capacitor

It forms the physical coupling link between transmission line and the terminal assemblies for the relaying of carrier signals. Its function is to provide high impedance to power frequency and low impedance to carrier signal frequencies.

Drain Coil

As shown in the figure the purpose of drain coil is to provide high impedance for carrier frequency and low impedance for power frequency.

Line tuner

It is connected in series with the coupling capacitor to form a resonant circuit or carrier signal frequency high pass filter or band pass filter. Its function is to match the impedance of the PLC terminal with the power line in order to impress the carrier frequency over the power line. In addition it also provides isolation from power frequency and transient overvoltage protection.

Line Trap or Wave Trap

It is a parallel L-C tank filter or band-stop filter connected in series with the transmission line. It presents high impedance to carrier signal frequencies and very low impedance to the power frequency.

Coaxial Cables: The coaxial cable is one type of electrical cable that carries high-frequency signals with low losses.

Protective Device: This device is used to protect the wave trap or line trap from damages.

Applications

- Transmission and distribution network
- Home control and automation
- Entertainment
- Telecommunication
- Security systems
- Automatic meter reading
- Telemetry
- Telephony
- Protective relaying

Limitations

- It requires a high signal to noise ratio
- The power line communication is not secure

Advantages

- Complex
- Reliability
- Cost-effective
- Lower attenuation